

Finance Tracker Intel v3 — Personal Investment Committee + Opportunity Radar

Draft 2 — Maximum-Depth Architecture & Product Specification

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Pre-flight: Foundational assumptions and product contract

Before any architecture, the contract that v3 must enforce end-to-end:

1. **Intel decides WHAT deserves capital, and WHY.** Deploy decides HOW MUCH and WHEN. Intel never owns allocation math; Deploy never invents thesis text. The boundary is a single typed handoff payload (defined later).
 2. **Numbers are deterministic, claims are sourced.** Anything that looks like a number on a card must trace to a deterministic source row with `value`, `source`, `ts_published`, `source_url`, `freshness_band`, `confidence`. LLMs may quote, summarize, debate, and synthesize — they must not invent metrics, scores, financials, price targets, or allocations.
 3. **Suppression is a first-class output.** "We don't have this signal yet" is a real, visible state, not a degraded BUY. Per-axis suppression replaces global `insufficient_data` flattening.
 4. **Visible action set is fixed.** Held tickers: BUY / HOLD / TRIM / SELL. Non-held opportunities: WATCH / BUY (if cleared standalone *and* portfolio-fit allows) / AVOID (only with sourced disqualifying flag). No posture buckets in UI. No broker-style overconfidence. No famous-investor cosplay.
 5. **Frontend stays clean and beginner-friendly.** No raw metric keys, no advanced jargon, no business-read wall, no analyst report dump. All complex analysis lives in the backend.
 6. **The user's portfolio is the reference frame.** Every recommendation is portfolio-aware (concentration, sector, factor, ETF overlap, role, tax-lots in the future). A "Buy" on a name we already have 9% of is not a Buy.
 7. **Markets are probabilistic.** No precision price targets. No "will go to \$185". Scenarios with explicit assumptions are allowed and must be labeled "scenario, not target". Confidence is enum {LOW, MEDIUM, HIGH}, never a numeric percentage.
 8. **Personal scope is fine.** This is one investor, ~34 holdings, periodic deposits, single-tenant. We optimize for decision quality per unit of evidence, not for fund-grade compliance. But we maintain audit trails and reproducibility because future-you will want them.
 9. **Run cadence is tiered.** Daily light refresh (deterministic only), weekly full radar pass, on-demand single-ticker committee. No "every keystroke" recompute. No "Run Agents" as the only refresh path.
 10. **The architecture must remain additive.** A failing module degrades, never crashes; a missing provider suppresses, never silences. Every layer can be turned off and the system still produces honest output.
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A. Draft 1 adversarial critique

Draft 1 was directionally correct. It is also full of things I would attack if I were a senior reviewer. The honest list of what is too shallow, missing, or wrong:

A.1 Too shallow

- **Decision policy was sketched, not specified.** I gave priority rules but did not enumerate the full state space. A real policy needs a machine-checkable table over (EQ, AT, PR, FIT, RT, TECH, CAT, REG, asset_class) with explicit transitions, including how a borderline case gets a WATCH instead of a HOLD, and how a thesis-broken signal upgrades to SELL even with thin data.
- **Source-pack format was a bullet list, not a schema.** Without a precise typed schema, the validator cannot reject hallucinations. Draft 2 includes a JSON schema with required fields, ID resolution rules, and enforced citation grammar.
- **Opportunity Radar did not specify ranking math.** "Pareto-ranked across four axes" is fine as a slogan; the actual algorithm needs tie-breakers, deduplication rules, suppression cascades, and per-screen quotas.
- **No backtesting plan.** A system that produces decisions without a way to measure them post hoc is just an opinion generator. Draft 2 has a full feedback/backtesting layer.
- **Cost discipline was hand-wavy.** "\$0.10–\$0.40 per run" is not a control system. Draft 2 specifies token budgets per persona, per tier, and a concrete cost cap algorithm.

A.2 Missing

- **No Investment Policy Statement (IPS) layer.** CFA-style portfolio management starts with the investor's own constitution: goals, risk tolerance, constraints, sell discipline, override rules. Without it, every "BUY" is unanchored. Draft 2 adds Layer 0.
- **No earnings-quality / forensic accounting module.** Draft 1 mentioned dilution and filing risk but lacked a Beneish-M-style earnings-quality probe, a cash-conversion test, and a related-party transactions check. These are exactly the things short-sellers find first.
- **No earnings-call / transcript intelligence.** Tools like Aiera have proven that transcripts contain forward-looking color (capex, demand, margin guidance) that filings miss. Even without paying for a transcript provider, the system should track surprises and revisions.
- **No sector / theme / factor exposure attribution.** Hedge funds learn the hard way that "diversified" is often correlated. Draft 1 mentioned ETF overlap; Draft 2 adds full factor and theme exposure attribution against the held portfolio.
- **No "thesis tracker" with explicit upgrade/downgrade conditions.** A real PM writes the thesis down and revisits it. Draft 2 adds a per-position thesis card with `must_be_true[]`, `would_change_view[]`, and a delta engine that flags drift.
- **No tax-aware turnover guardrail.** Even for a personal account, "Sell because price is full" without considering holding period, lot basis, and wash-sale risk is bad advice. Draft 2 introduces a tax-aware modifier (deferred to phase 2 PRs).

- **No options/short-interest awareness.** I deferred options indefinitely. That was lazy. Implied volatility skew and short interest are cheap, asymmetric signals if read correctly. Draft 2 includes them as a tertiary "options/short tape" module with strict suppression.
- **No regime-aware tilts.** The macro module just emitted a regime label. Draft 2 specifies how the regime modulates conviction and capital priority (not stock picks).
- **No behavioral-finance safeguards.** A personal investment committee that doesn't actively counteract overconfidence, anchoring, loss aversion, and disposition effect will exhibit them. Draft 2 adds an explicit behavioral guardrails module that flags risky user actions ("you've been adding to a losing position 3 runs in a row — here's the bear case again").
- **No event/catalyst-driven mode.** Catalyst investing is a real edge for retail. Draft 1 listed `catalyst_watch_v1` but never integrated it into ranking or alerting.
- **No replay / paper portfolio.** Without replay we can't validate the engine on history.
- **No alerting layer.** A committee that runs weekly without surfacing "something just changed" is half-built. Draft 2 adds a delta engine + alerting.
- **No "do not act" output.** Sometimes the right answer is nothing changed; do not act. The system needs to express that with confidence so the user resists overtrading.

A.3 What a top hedge fund analyst would attack

- "Your `quality_moat_v1` is just ROIC + margin. Where is reinvestment runway? Where is unit economics? Where is competitive position over 5–10 years?" → Add reinvestment-runway and structural-position dimensions.
- "Your bear case is generated by the same provider as your bull case. That is not a real red team." → Force the bear and red-team prompts to use a different model (or temperature), and require external short-seller-style rubric coverage.
- "Your valuation is multi-metric but never compares to peers, only to history." → Add peer-set valuation context (with explicit peer-set construction rules and conflicts disclosed).
- "You don't track estimate revisions. Revisions are the most predictive single signal in equities." → Add an estimate-revisions module (consensus EPS/revenue revisions, surprise history). Optional in free tier; recommended for paid tier.
- "You don't have an event/catalyst clock that ties to thesis check-ins." → Wire `catalyst_watch_v1` into the thesis-tracker and alerting layer.

A.4 What a wealth manager would attack

- "Where is the Investment Policy Statement?" → Layer 0 added.
- "Where is the asset allocation policy independent of stock picks?" → Add `policy_v1` that tracks target sleeves (US large/SMID, Intl, FI, Cash, Alts) and surfaces drift.
- "Where is the deposit cadence and rebalancing rhythm?" → Cadence policy and Deploy handoff schema include this.
- "Why are you only thinking about equities and ETFs? Fixed income, gold, crypto, cash deserve a slot." → Asset-class router (`fund_intel_v1` extended) covers this; FI is a known gap until Phase 2.
- "Where is the tax efficiency story?" → Future-phase module; explicit deferment with rationale.

A.5 What a quant / factor researcher would attack

- "You list factors but don't compute exposures." → Add `factor_exposure_v1` that estimates portfolio-level exposures to value/quality/momentum/size/low-vol via either ETF passthrough or bottom-up regression on held names.
- "Your momentum module is a label, not a factor." → Add multi-horizon momentum (20/60/125/250d) à la CTA literature with risk-parity weighting at signal level.
- "Your scores collapse on insufficient data. Sparse imputation should be possible without lying." → The decision policy must operate on *bands* (ordinals) so missing axes do not algebraically zero out neighbors.
- "Where is volatility-targeting in conviction?" → Add a volatility-aware conviction modifier (high-vol names get capped MEDIUM unless other axes are very strong).

A.6 What a forensic accountant would attack

- "No Beneish M-Score, no Sloan accruals score, no cash-conversion ratio, no earnings-vs-cashflow divergence detector." → Add `earnings_quality_v1`.
- "No related-party transactions or auditor-change scan." → Add to `filing_risk_v1`.
- "No restatement detection or 8-K item tracking." → Add to `filing_risk_v1`.
- "No going-concern / debt-maturity wall analysis." → Add to `filing_risk_v1`.
- "No SBC-as-percent-of-FCF tracking; that is the single biggest distortion in modern equity analysis." → Add to `dilution_risk_v1`.

A.7 What a risk manager would attack

- "No VaR, no stress test, no scenario engine on the portfolio." → Add `risk_monitor_v1` that runs at least: 2008 replay, 2020 replay, 2022 replay, +rate-shock, +USD-shock, +oil-shock against current weights.
- "No correlation/clustering analysis." → Add factor + sector clustering on current holdings; flag hidden concentration.
- "No drawdown discipline / drawdown-state-aware behavior." → Add a portfolio-level drawdown monitor that escalates conservatism when drawdown thresholds are crossed.
- "No position-size limits enforced bidirectionally." → `portfolio_governor_v1` enforces both lower (de-minimis) and upper (concentration cap, sector cap) bounds.

A.8 What a behavioral finance expert would attack

- "Disposition effect is unaddressed. Investors hold losers too long and sell winners too early. Your engine will replicate this if it just summarizes. Where is the active counter?" → Add behavioral guardrails: when a winner triggers TRIM, prompt second-look; when a loser triggers BUY-more, flag if no thesis change.
- "Anchoring on cost basis: do not show cost basis as a primary decision variable. Show fundamentals first." → UI rule: cost basis is in the drawer, not on the card.
- "Recency bias: weekly committee runs will overreact to the latest macro print. Force a multi-horizon view." → Decisions should reference a longer-window signal (90/180d) before a shorter-window one.

- "Herding: do not surface 13F flows as 'famous investor X bought this' top-line." → Use 13F as evidence, not as cosplay.
- "Confirmation bias: bull and bear must be generated independently and shown side by side, not synthesized into a soft consensus before the user sees the disagreement." → Drawer shows Bull and Bear separately and surfaces dissent.

A.9 What a software architect would attack

- "Your snapshot store is JSONB blobs without schema versioning. Migrations will be painful." → Snapshot rows carry `schema_version`; readers handle older versions; migrations are explicit.
- "Your typed contract `IntelToDeployHandoff` is mentioned but not defined." → Defined precisely in this draft.
- "You have no event bus or job queue." → For now, FastAPI BackgroundTasks + Supabase as the queue is fine; if scale grows, switch to a proper queue. Specified explicitly in Layer L.
- "Your test strategy is per-module, not contract-level." → Add contract tests that pin the `IntelCardV3` schema, decision-policy outputs on a frozen set of source packs, and `Intel→Deploy` handoff stability.
- "Re-use of `agent_runs.allocation` JSONB for two purposes (Deploy data + Intel reasoning) is technical debt." → Migration plan separates these into distinct tables.

A.10 What would fail in production

1. **EDGAR rate-limit bursts.** SEC enforces 10 req/s globally per IP. A naive radar scan over 200 names will hit this. Mitigation: token-bucket with 6 req/s sustained, retry/backoff, daily cron pre-warm.
2. **yfinance 429s and silent NaNs.** yfinance frequently returns NaN with no error. Mitigation: every numeric value passes through a `coerce_or_suppress` sanitizer that emits suppressed rows on NaN/None/inf.
3. **LLM provider outages mid-run.** Mitigation: snapshot writes are checkpointed per ticker; if committee fails, the deterministic decision still produces an action; `committee_present=false` flagged in UI.
4. **LLM hallucinated tickers in radar.** The single biggest risk. Mitigation: closed universe — radar suggestions must come from `universe_v1`; out-of-universe = rejected by validator.
5. **Drift between deterministic action and LLM narrative.** Mitigation: decision policy is the only producer of action; LLM output is rendered as Bull/Bear/Risk arguments only; there is no LLM-derived `action` field anywhere downstream.
6. **Page-load vs run inconsistency reappears.** Mitigation: snapshots are per-run, content-hash addressed, immutable; cards always render from one snapshot id; no fallback blending.
7. **Posture-bucket regressions.** Mitigation: regression tests pin the visible filter contract; PR template forbids new label categories without a spec change.
8. **Cost runaway from radar + committee on 200-name universe.** Mitigation: deterministic ranking gates LLM calls to top-N; per-run token cap with graceful abort.
9. **Stale snapshot served indefinitely.** Mitigation: `fresh_until` per fact + per-snapshot; UI shows "as of" badge; user can force refresh.
10. **Provider schema drift.** Mitigation: each adapter has a `provider_health` ping that runs daily; schema-drift alerts fail loudly.

B. Research synthesis — what the best investors and tools actually do

This section distills durable principles from professional practice. We will not blindly copy any tool or manager; we extract patterns and translate them to a single-tenant personal app.

B.1 Quality investing (Buffett / Smith / Train)

- **Core idea:** Buy businesses whose ROIC sustainably exceeds WACC, with structural moats (network effects, switching costs, scale, brand/IP), and hold them long. Terry Smith's Fundsmith uses ROCE as the primary lens; aggregate ROCE in the high 20s–30s vs the FTSE 100 $\approx$ 17%.
- **Useful for v3:** A `quality_moat_v1` that publishes a band only when $\text{ROIC} > \text{cost-of-capital proxy}$ for $\geq 3y$; a `capital_allocation_quality_v1` that scores buyback timing, dilution, and value-accretive M&A.
- **Limit:** Quality often looks expensive on traditional multiples. The decision policy must not use single-PE labels.

B.2 Value investing (Graham / Klarman / Greenblatt)

- **Core idea:** Margin of safety. Buy when price $<$ intrinsic value with a meaningful cushion. Sell when the gap closes — Klarman's rule of selling at \$18 when worth \$20, "never the last nickel."
- **Greenblatt's Magic Formula** explicitly combines value (earnings yield = EBIT/EV) with quality ($\text{ROC} = \text{EBIT}/\text{invested capital}$). This is the foundation we should adapt: `valuation_context_v1` $\times$ `quality_moat_v1` jointly, not separately.
- **Useful for v3:** Sell discipline is explicit: when valuation is full or expensive AND attractiveness has not improved, the policy must consider TRIM. Thesis-creep detection (Klarman): when our view of a name has drifted upward without new facts, the system flags it.
- **Limit:** Pure value can be a value trap. Always pair with quality and forensic accounting.

B.3 GARP and quality-growth (Lynch / Train)

- **Core idea:** Growth at a reasonable price. PEG with quality filter. Reinvestment runway matters more than peak margins.
- **Useful for v3:** `return_drivers_v1` decomposes expected return into revenue growth, margin trajectory, multiple change, and capital return. If "growth" carries 60% of expected return, the gate becomes "is the growth durable" not "is it cheap".
- **Limit:** Growth analysis collapses without forward visibility. Suppress when multi-year history is missing.

B.4 Momentum and trend following (AQR / CTA literature)

- **Core idea:** Price persistence is broad, global, and durable across 150+ years and 46 countries. Multi-horizon trend following (20/60/125/250d) with risk-parity at the signal level reduces single-horizon fragility.
- **Useful for v3:** `technical_confirmation_v1` becomes a multi-horizon momentum module that confirms or denies a fundamental thesis. It is never the primary driver. Combined with quality, it filters value traps.
- **Limit:** Momentum has crash risk; mitigate by reducing exposure during volatility spikes.

B.5 Factor investing (Fama-French / MSCI / JPM)

- **Core idea:** Multi-factor exposure (value, quality, momentum, size, low-vol) reduces dependence on any single style. 2025: momentum extended outperformance; quality lagged; value rebounded. Cycle awareness matters.
- **Useful for v3:** `factor_exposure_v1` estimates portfolio-level factor loadings. The system flags hidden concentration ("you're 70% momentum-tilted via your tech holdings"). Decision policy includes a "would this trade increase or decrease my factor concentration?" check.
- **Limit:** Estimating factor exposure on a 34-name personal book is noisy; treat as informational, not a hard gate.

B.6 Risk management (Marks / Resonanz / hedge fund practice)

- **Core idea:** Risk is the probability of loss, not volatility. Asymmetry — capping downside while preserving upside — is the cornerstone of superior investing. Position sizing in isolation hides correlated exposures; sector and factor caps are essential. Stress tests against historical crises (2008, 2020, 2022) inform leverage and hedging.
- **Useful for v3:** `risk_monitor_v1` runs scenario replays on the current portfolio. `portfolio_governor_v1` enforces position, sector, factor, and cluster caps simultaneously. Conviction is downgraded under stress.
- **Limit:** VaR/CVaR are abused tools; we use them as one of many lenses, not the lens.

B.7 Forensic accounting / short-seller research (CFRA / Beneish / Sloan)

- **Core idea:** Earnings quality is the single highest-yield diagnostic short-sellers use. Cash-conversion ratio (OCF / NI) consistently below 0.8 is a flag. Receivables growing faster than sales is a flag. Inventory days rising while revenue grows is a flag. Beneish M-score < -1.78 is "low-manipulation". Short interest rises 19 months *before* misconduct disclosure on average.
- **Useful for v3:** `earnings_quality_v1` (new) computes Beneish M, Sloan accruals, cash-conversion, DSO trend, inventory-days trend. `filing_risk_v1` flags going-concern, late filings, auditor changes, restatements, RPTs above threshold. `dilution_risk_v1` tracks SBC as % of FCF — currently the largest distortion in modern equity analysis.
- **Limit:** All of these are lagged signals. Pair with technical/short-interest divergence.

B.8 Activist / event-driven / catalyst (Ackman / Loeb / event-funds)

- **Core idea:** Catalysts (earnings, regulatory, M&A, capital structure events) create asymmetric setups. Insider buying clusters and 13F accumulation by quality holders are useful corroborating signals — never primary.
- **Useful for v3:** `catalyst_watch_v1` tracks scheduled catalysts (earnings, ex-div, regulatory windows). `insider_inst_v1` (new) tracks Form 4 insider buys with significance filters and 13F changes from a curated list of "respected holders". Both feed Opportunity Radar's catalyst and accumulation screens.
- **Limit:** Event windows often disappoint. Use as a scoring modifier, not a primary BUY trigger.

B.9 Macro / regime-aware investing (Bridgewater / risk-parity)

- **Core idea:** Regime (growth, inflation, rates, credit, USD) tilts which factors and sectors work. CTAs use multi-horizon trend with regime detection to switch sleeves. We will not replicate this; we will use macro to *modulate conviction*.

- **Useful for v3:** `macro_regime_v1` reads FRED series (10Y, 2Y, fed funds, U-rate, ISM proxies, USD, HY spread) and emits one regime label + risk-on/risk-off tilt per run. The decision policy uses regime as a conviction modifier and as input to "why now / why not now" copy.

B.10 Behavioral finance (Kahneman / Thaler / CFA)

- **Core idea:** Overconfidence inflates trading; loss aversion holds losers too long; anchoring fixates on cost basis; disposition effect sells winners too early.
- **Useful for v3:** Behavioral guardrails (`behavioral_guardrails_v1`): flag adding-to-loser without thesis change; flag selling-winner-without-fundamental-change; suppress cost basis from primary card; require the user to accept the bear case before any "BUY more" of a position they already own.

B.11 Wealth manager / IPS practice (CFA Institute)

- **Core idea:** The Investment Policy Statement is the constitution. It defines goals, risk tolerance, time horizon, liquidity needs, tax constraints, asset allocation policy, and review cadence. Nothing the engine recommends should violate it.
- **Useful for v3:** Layer 0 (User Investment Constitution) becomes the precondition for every recommendation. The decision policy reads from it. Violations are blocked or flagged.

B.12 Performance attribution (Brinson-Fachler / multi-factor)

- **Core idea:** Decompose returns into allocation effect, selection effect, and (multi-factor) systematic vs idiosyncratic contributions. Acts as a feedback loop on whether the strategy's alpha came from where the manager claimed.
- **Useful for v3:** `attribution_v1` (Phase 2) attributes realized returns vs benchmark across allocation/selection/factors. Decision-quality scoring (the system is "right" if the action it recommended outperformed the alternative on the relevant horizon).

B.13 Best-in-class product patterns (institutional terminals / AI copilots)

- **Bloomberg, FactSet, Refinitiv:** dominant for research; multi-platform setups pair a terminal with web monitoring (IR, filings, press, product). Bloomberg added natural-language "AskB" agent over its document corpus. Cost: \$30k+/seat/year.
- **Aiera:** earnings call transcripts + AI summaries; partnered with Anthropic models (Claude 3 Opus / 3.5 Sonnet) for summarization; recently added expert-call transcripts via Third Bridge.
- **Atom Finance:** consumer-grade institutional features (verified chat, X-Ray, real-time portfolio); illustrative of the retail aspiration. Acquired in 2024, ceased normal operations late 2025.
- **Robo-advisors (Betterment / Wealthfront):** clean UX, IPS-driven, tax-loss harvesting, simple allocation; but no thesis intelligence.
- **What to copy:** clean cards, plain-English summaries, citation-first display, side-by-side bull/bear, "what changed" deltas, multi-horizon view.
- **What not to copy:** dense terminals, raw metric overload, real-time alert spam, "famous investor" cosplay rails, social-feed noise.

B.14 LLM grounding / RAG best practices (industry research)

- **Core idea:** LLMs hallucinate up to 41% in finance queries; mitigation requires multi-layered controls: structured prompt design, RAG with verifiable evidence, targeted fine-tuning aligned to domain truth. Provenance-aware pipelines that expose uncertainty are non-negotiable.
 - **Useful for v3:** Source-pack-only inputs; citation grammar enforced; numeric mentions resolve to source-pack values within tolerance; per-claim provenance; out-of-universe rejection.
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C. Master alpha map

This is the explicit list of every plausible source of personal-app edge. For each: *why it can work*, *why it can fail*, *data required*, *module required*, *UI expression*, *Deploy handoff signal*. This is the closest thing to a "menu of edges" we have.

C.1 Better data hygiene (citation-first, freshness-aware)

- **Why it works:** Most retail tools display numbers without timestamps or sources. We always show source + asof. The user can audit. The model has fewer attack surfaces for hallucination.
- **Why it fails:** Free providers drift; conflicts go unflagged.
- **Data:** Adapter health checks; reconciliation logic; per-fact TTL.
- **Module:** `data_truth_v1`, `snapshot_store_v2`, `provider_health_v1`.
- **UI:** "as of" badge; click-to-citation.
- **Deploy signal:** `data_quality_band` per ticker; Deploy can refuse to deploy on thin if user opts-in.

C.2 Faster synthesis (multi-source source-pack + LLM committee)

- **Why it works:** Reading a 10-K + transcript + estimate revisions + price action + ETF overlap is hours of work; an LLM committee over a curated source-pack takes seconds.
- **Why it fails:** LLM hallucination; bias toward narrative.
- **Data:** Filings (SEC), transcripts (paid in future tier), prices, fundamentals.
- **Module:** `source_pack_builder`, `committee_v1`, `committee_validator`.
- **UI:** Drawer Bull/Bear bullets with citations.
- **Deploy:** None directly; conviction band feeds Deploy.

C.3 Source-backed thesis tracking

- **Why it works:** We force every position to have a written thesis with `must_be_true[]`. Drift is detected.
- **Why it fails:** User doesn't update theses; bot-generated theses lack judgment.
- **Data:** Thesis store, fundamental snapshots, time series.
- **Module:** `thesis_tracker_v1`.

- **UI:** "Thesis check" pill on card; "Drift detected" in What-Changed.
- **Deploy:** `thesis_status` $\in$ {intact, drifting, broken} feeds capital priority.

C.4 Red-team discipline

- **Why it works:** A separate prompt with a different temperature/model that *only* searches for reasons not to own. This is what short-seller research is.
- **Why it fails:** Confirmation bias if bull and bear share context.
- **Data:** Forensic accounting modules; filings; short interest.
- **Module:** `red_team_v1`, `earnings_quality_v1`, `filing_risk_v1`, `dilution_risk_v1`.
- **UI:** "What could go wrong" bullets, citations.
- **Deploy:** Critical sourced flag $\rightarrow$ Deploy capital priority floor.

C.5 Valuation discipline (multi-metric, peers + history)

- **Why it works:** Single-PE labels are wrong half the time. We compare to history and peers across multiple metrics.
- **Why it fails:** Peer set is contestable; history may not be relevant in regime change.
- **Data:** Multiples, peer set, sector/factor benchmarks.
- **Module:** `valuation_context_v1`, `peer_set_v1`.
- **UI:** "Cheap on 4 of 6 measures vs 5y history; sector context: ...".
- **Deploy:** `valuation_band` feeds capital priority.

C.6 Factor awareness

- **Why it works:** Hidden factor concentration is the most common portfolio mistake.
- **Why it fails:** Estimation noise on small books.
- **Data:** Factor returns proxies, holdings, ETF passthrough.
- **Module:** `factor_exposure_v1`.
- **UI:** Portfolio Command Center "factor mix" chip.
- **Deploy:** Constraint when capital priority would worsen factor concentration.

C.7 Portfolio-fit awareness

- **Why it works:** A "BUY" on a 9% position is not the same as a BUY on a 0.5% position. Fit is the difference.
- **Why it fails:** Requires accurate position sizing and target weights.
- **Data:** Holdings, target sleeves, IPS.
- **Module:** `portfolio_governor_v1`.
- **UI:** "How it fits my portfolio" drawer section.

- **Deploy:** capital_priority, max_position_pct, must_not_exceed_pct.

C.8 Avoiding concentration mistakes

- **Why it works:** Caps prevent emotional doubling-down on a winner that has already become 12%.
- **Why it fails:** Rigid caps misfire on intentional concentration.
- **Data:** IPS caps, current weights.
- **Module:** portfolio_governor_v1.
- **UI:** "At cap" / "Near cap" badge.
- **Deploy:** Hard must_not_exceed_pct.

C.9 Sell / trim discipline

- **Why it works:** Klarman-style: sell at 18 when worth 20. Greenblatt-style: rotate when better mispricing exists. Pre-committed rules beat in-the-moment judgment.
- **Why it fails:** Tax inefficiency, narrative attachment.
- **Data:** Valuation context, thesis status, fit, alternatives.
- **Module:** decision_policy_v1 TRIM rules; behavioral_guardrails_v1 for second-look.
- **UI:** "Trim/Sell desk" section with reason codes.
- **Deploy:** capital_priority $\in$ {trim_first, sell_first}.

C.10 Spotting thesis drift

- **Why it works:** Theses creep upward as price rises; system catches it.
- **Why it fails:** Requires periodic user thesis review.
- **Data:** Thesis store, fundamentals time series.
- **Module:** thesis_tracker_v1.
- **UI:** "Drift detected" delta.
- **Deploy:** thesis_status.

C.11 Catching filing red flags early

- **Why it works:** Going-concern, restatements, late filings, auditor changes are leading indicators.
- **Why it fails:** False positives on legitimate filings.
- **Data:** EDGAR full-text + XBRL.
- **Module:** filing_risk_v1.
- **UI:** "Filing flag: serious" with citation.
- **Deploy:** Veto.

C.12 Earnings quality scrutiny (Beneish, Sloan, cash conversion)

- **Why it works:** Earnings manipulation is detectable in ratios long before disclosure.
- **Why it fails:** Some industries legitimately diverge (financials, real estate).
- **Data:** Multi-quarter financials.
- **Module:** earnings_quality_v1.
- **UI:** "Earnings quality: clean / mixed / flagged" with one-line why.
- **Deploy:** Veto if flagged.

C.13 Finding opportunities outside current holdings

- **Why it works:** A personal book of 34 names is small; the universe is huge; many high-quality names go unconsidered for psychological reasons (unfamiliar, recent IPO, foreign).
- **Why it fails:** Universe expansion → cost; LLM hallucination of tickers.
- **Data:** Closed universe (S&P 1500, Russell 1000, sector ETFs, curated quality list, watchlist).
- **Module:** opportunity_radar_v1, universe_v1, all axis modules.
- **UI:** "New opportunities" section with screen tag and 1-line why.
- **Deploy:** WATCH/BUY signals on non-held names, only when standalone bar met.

C.14 Using personal portfolio context better than generic tools

- **Why it works:** Generic screeners don't know my book; my Buys are filtered through *my* concentration, sector tilts, ETF overlap, and policy.
- **Why it fails:** Requires accurate holdings and IPS.
- **Data:** Holdings, IPS.
- **Module:** portfolio_governor_v1, policy_v1.
- **UI:** Card "fit" chip; portfolio-aware ranking.
- **Deploy:** Per-ticker capital priority is portfolio-aware.

C.15 Tracking what changed since last run

- **Why it works:** Most decisions are about deltas, not absolutes. Action upgraded? Red flag added? Thesis broken?
- **Why it fails:** Noise from data revisions.
- **Data:** Snapshot diffs.
- **Module:** delta_engine_v1.
- **UI:** "What Changed" section.
- **Deploy:** recent_delta in handoff for prioritization.

C.16 Post-decision feedback loop

- **Why it works:** Without measurement, the system cannot improve and the user cannot calibrate trust.
- **Why it fails:** Short windows are noise; long windows lose engagement.
- **Data:** Decision logs, prices, benchmarks.
- **Module:** feedback_v1 (Brinson-style attribution at 7/30/90/180/365d windows; confusion matrix on actions vs subsequent returns).
- **UI:** "How was the system on this name?" in drawer; portfolio-level attribution in a separate Performance tab (Phase 2).
- **Deploy:** None directly; refines conviction calibration.

C.17 Avoiding behavioral mistakes

- **Why it works:** Pre-committed rules beat in-the-moment urges.
- **Why it fails:** Rigid rules reduce optionality.
- **Data:** User actions, holding periods, prior decisions.
- **Module:** behavioral_guardrails_v1.
- **UI:** "Second look" prompt; bear-case-acknowledged checkpoint before high-conviction moves.
- **Deploy:** Soft block / warning UX before user clicks "Use AI plan".

C.18 Knowing when not to act

- **Why it works:** Most days the right answer is do nothing. Inactivity is a virtue (Smith).
 - **Why it fails:** Engagement metrics push toward action.
 - **Data:** Delta engine + decision policy.
 - **Module:** decision_policy_v1 HOLD with do_not_act=true.
 - **UI:** Empty Best-Buys with affirmative copy: "No new high-conviction action this run."
 - **Deploy:** Deploy can default to wait mode.
-

D. Best-possible architecture

The architecture is a 9-layer system. Every layer has a typed contract; layers do not skip.

```

Layer 8 Feedback / Attribution / Replay
Layer 7 Frontend Product Contract (Next.js)
Layer 6 Decision Policy Engine
Layer 5 LLM Investment Committee (multi-agent, validated)
Layer 4 Opportunity Radar (closed universe + screens)
Layer 3 Portfolio Governor (book-aware constraints)
Layer 2 Company / Fund Intelligence Modules (axes)
Layer 1 Data Truth Layer + Snapshot Store
Layer 0 User Investment Constitution (IPS)

```

Layer 0 — User Investment Constitution (policy_v1)

This is the foundation. Without it, every recommendation is unanchored. Persisted in Supabase, editable in a Settings UI (Phase 2 PR — initial PR seeds defaults).

Contents:

- **Goals:** long-term growth (default), with optional sub-goals (income, capital preservation, tax-efficient compounding).
- **Time horizon:** primary (10y+), secondary if relevant.
- **Risk tolerance:** ordinal {low, moderate, moderate-high, high}; mapped to portfolio-level vol band and drawdown tolerance.
- **Liquidity needs:** target cash floor, deposit cadence, withdrawal expectations.
- **Asset allocation policy (target sleeves):** US Large/SMID, Intl Dev, Intl EM, Fixed Income, Cash, Alts (incl. crypto), Speculative bucket.
- **Concentration limits:** per-position cap (default 8%), sector cap (default 30%), factor cap (default 60% on a single factor), speculative bucket cap (default 5%).
- **Preferred holding style:** quality-compounder bias (default), with allowance for GARP and value rotations.
- **Sell discipline:**
 - Sell when thesis broken (sourced).
 - Trim when at concentration cap (regardless of view).
 - Trim when valuation expensive AND attractiveness not strong.
 - Trim/Sell when better-fit alternative exists with higher expected risk-adjusted return AND policy allows turnover.
- **Engine override rules:**
 - User can override any TRIM/SELL with explicit reason; logged.
 - User cannot override an AVOID veto without a "I understand the disqualifying flag" acknowledgement.
- **Engine never allowed to:**
 - Auto-execute trades.
 - Generate price targets.
 - Recommend speculative names above the speculative cap.
 - Recommend turnover above policy turnover ceiling.
 - Surface raw metric keys.
 - Make claims without sources.
 - Imply certainty.

Outputs to other layers: A Policy struct that every other layer reads. Changes to policy are versioned; decision history records which policy version was active.

Layer 1 — Data Truth Layer + Snapshot Store

Already detailed in module section. Key responsibilities:

- **Source registry:** Each provider has a config row with priority, TTL per fact, schema version, health endpoint.
- **Source snapshots:** Per-fact rows with value, unit, ts_observed, ts_published, source, source_url, freshness_band, confidence.
- **Timestamping:** Both observed (when we saw it) and published (when source dated it).
- **Reconciliation:** When two sources disagree by > tolerance, prefer higher priority and log conflict.
- **Stale-data handling:** freshness_band ∈ {fresh, stale, expired}; expired = suppressed.
- **Confidence scoring:** Per-fact, decreases with TTL distance and increases with cross-source agreement.
- **Metric normalization:** Units, currencies, fiscal-year alignment.
- **Filing extraction:** EDGAR full-text + XBRL with deterministic flag extractors (regex for going-concern, restatement, auditor change, RPT thresholds).
- **Source citation store:** Every fact carries its citation; UI hover reveals.
- **No-source/no-claim enforcement:** Validators reject any value without citation.
- **Audit trail:** Append-only log of all writes, with user/system actor.

Layer 2 — Company / Fund Intelligence Modules

Each module is a deterministic axis with publish/suppress semantics.

Module	Purpose	Publish gate
business_model_v1	Revenue mix, segment breakdown	Need ≥1y segment data
return_drivers_v1	Decompose expected return: revenue, margin, multiple, capital return	≥2y history
quality_moat_v1	ROIC vs WACC, moat signals, durability	≥3 quality inputs
management_quality_v1	Tenure, capital allocation track record, insider buys	Optional
capital_allocation_quality_v1	Buyback timing, M&A accretion, dividend record	≥3y
earnings_quality_v1	Beneish-M, Sloan accruals, cash conversion, DSO/DPO trends	≥4 quarters
valuation_context_v1	Multi-metric vs history + peers	≥3 metrics OR ≥2 baselines
growth_durability_v1	Reinvestment runway, organic vs acquisitive growth	≥3y revenue + ≥2y unit-econ proxies
margin_trajectory_v1	GM/OM/FCFM trends, peak vs trough margin	≥3y
balance_sheet_resilience_v1	Net debt/EBITDA, interest coverage, debt maturity wall	≥1 quarter detailed BS
cashflow_quality_v1	OCF vs NI, FCF vs OCF, working capital trend	≥4 quarters
dilution_risk_v1	Share count trajectory, SBC% of FCF, ATM/convertibles	≥2y share counts
filing_risk_v1	Going-concern, restatement, auditor change, RPT, late filings	EDGAR coverage required

Module	Purpose	Publish gate
customer_concentration_v1	Top customer % of revenue if disclosed	Disclosure-dependent
regulatory_litigation_risk_v1	Active material litigation, regulatory actions	EDGAR + news
geopolitical_supply_chain_v1	Geographic revenue mix, sanction exposure	Disclosure-dependent
catalyst_watch_v1	Upcoming earnings, ex-div, regulatory windows	Calendar-dependent
technical_confirmation_v1	Multi-horizon momentum (20/60/125/250d), RSI, drawdown, RS-vs-SPY	≥6mo price history
sentiment_noise_filter_v1	News tone with source-quality weighting; social demoted	Source count threshold
short_options_tape_v1	Short interest, days-to-cover, implied vol skew	Provider-dependent
factor_exposure_v1	Per-name and portfolio factor loadings	Universe-aware
fund_intel_v1	ETF/fund role, holdings, overlap, expense, replication	Holdings file present
peer_set_v1	Constructed peer sets for relative valuation	Sector/industry mapping
insider_inst_v1	Form 4 insider buy clusters; 13F changes from curated holders	EDGAR coverage
tax_turnover_v1 (Phase 2)	Holding periods, lot basis (when added), wash-sale guard	User-data dependent

Each module emits:

```
ModuleOutput = {
  module: "...",
  ticker: "...",
  band: <ordinal enum specific to module>,
  signals: [{label, value, citations: [fact_id]}],
  inputs_used: [...],
  inputs_missing: [...],
  data_quality: {coverage_pct, freshness_band, confidence},
  suppressed: bool,
  suppression_reason: "..." | null,
  schema_version: "1.0"
}
```

Layer 3 — Portfolio Governor

Inputs: holdings, IPS, factor exposures, ETF overlap, concentration, sector exposure.

Outputs per ticker:

```
PortfolioFit = {
  ticker: ...,
  current_weight: float,
  target_weight: float | null,
  min_weight: float | null,
  max_weight: float,
  must_not_exceed_weight: float,
  fit_band: enum {underweight, on_target, overweight, breach, blocked},
```

```

capital_priority: enum {urgent_buy, opportunistic_buy, hold, trim_first, sell_first},
blockers: [str],    // e.g. ["sector_cap_tech_30%", "concentration_8%"]
rationale: str      // plain English
}

```

The governor enforces upper, lower, sector, factor, cluster, and asset-class caps simultaneously. It is the only layer that knows portfolio-wide context.

Layer 4 — Opportunity Radar

Closed universe; multiple parallel screens; deterministic ranking; LLM committee gated to top-N. (Detailed in Section G.)

Layer 5 — LLM Investment Committee

Multi-agent, source-pack-only, validator-enforced. (Detailed in Section H.)

Layer 6 — Decision Policy

The state machine that produces final action. (Detailed in Section I.)

Layer 7 — Frontend Product Contract

Clean, beginner-friendly, citation-first UI. (Detailed in Section J.)

Layer 8 — Feedback / Attribution / Replay

Decision logs + post-hoc performance + replay engine + behavioral metrics. (Detailed in Section K.)

E. Data / source / integration map

E.1 Source classes (free → paid)

Each row: name, what it enables, reliability, freshness, cost, licensing, hallucination risk, when, fallback.

Free / open

Source	Enables	Reliability	Freshness	Cost	Licensing	Hallucination risk	When	Fallback
SEC EDGAR (HTML + XBRL)	Filings, financial statements, share counts, restatements, auditor changes, RPTs, going concern, going-concern language, 13F, Form 4	High	Real-time-ish (sub-minute for submissions; under a minute for XBRL)	Free	Public domain; rate limit 10 req/s	Low when parsing deterministically	Now (Phase 0–1)	None — primary truth source for filings
FRED (St. Louis Fed)	Macro series (rates, inflation, employment, USD, ISM proxies)	High	Daily–monthly	Free with API key; 120 req/min	Public domain	Low	Now (Phase 1)	Hard-code key series if outage
yfinance (unofficial)	Prices, fundamentals, info, options chain	Medium	EOD reliable; intraday flaky	Free	Unofficial — terms ambiguous; risky for production at scale	Medium (silent NaN)	Now; carry forward into v3 with strict NaN sanitizer	Finnhub fundamentals
Finnhub (free tier)	News, earnings calendar, basic fundamentals	Medium	Daily	Free 60 req/min	Permissive personal	Low for news; medium for fundamentals	Now	yfinance
Polygon (free tier)	Aggregates (delayed)	Medium	Delayed 15 min on free	Free limited	Permissive	Low	Now	yfinance
CoinGecko	Crypto prices, market cap, circulating supply	Medium-High	Daily	Free	Permissive	Low	Now (already integrated)	None for crypto
ETF issuer holdings (Vanguard / iShares / State Street / Invesco)	ETF holdings JSON/CSV	High	Daily–monthly	Free	Permissive personal use	Low	Phase 1	EigenDex / ETF Research Center
EigenDex / ETF Research Center / Wise Sheets	ETF overlap analysis	High	Daily	Free	Personal use	Low	Phase 1 fallback	issuer JSONs

Source	Enables	Reliability	Freshness	Cost	Licensing	Hallucination risk	When	Fallback
Securities DB / Earnings Feed (free tiers)	Aggregated 13F + Form 4	Medium-High	Daily	Free with limits	Permissive	Low	Phase 2	EDGAR direct parse
SEC EDGAR full-text search	Going-concern keyword scan	High	Real-time-ish	Free	Public	Low	Phase 1	Per-filing parse

Affordable personal (paid future tier)

Source	Enables	Cost ballpark	Hallucination risk	When
Polygon paid	Real-time + better historicals	\$30–\$200/mo	Low	Phase 2 if needed
Tiingo	Fundamentals + historicals	\$10–\$50/mo	Low	Phase 2
Financial Modeling Prep (FMP)	Estimates, transcripts (limited), key metrics	\$15–\$80/mo	Medium	Phase 2 — replaces yfinance fundamentals
EOD Historical Data	Fundamentals, dividends, splits	\$20–\$80/mo	Low	Phase 2
Alpha Vantage	Fundamentals + technicals	Free + paid tiers	Medium	Phase 2
AlphaSense / Sentieo (academic / lighter plans)	Transcript search	\$\$\$ — likely too expensive personal	Low	Defer
Aiera / similar	Earnings call transcripts + summaries	\$\$\$ enterprise	Low	Defer indefinitely; not personal-tier

Premium / enterprise (do not buy)

- Bloomberg Terminal, FactSet, Refinitiv, S&P Capital IQ — \$20k–\$50k+/seat/year. Out of scope.

E.2 Per-source playbook

For each source we use, the integration pattern:

1. **Adapter module** (`adapters/source_x.py`) — only place that talks to the source.
2. **Schema lock** — pinned response schema with version; drift detection.
3. **Health check** — daily ping; alerts on schema drift or downtime.
4. **TTL per fact** — different TTLs for different fields (price=15min EOD, financials=24h, filings=7d, share counts=7d, macro=24h).
5. **Citation builder** — emits `{value, source, ts_published, source_url}` for every value.
6. **NaN/None sanitizer** — coerce-or-suppress.
7. **Conflict reporter** — when two sources resolve the same fact differently.

8. **Backoff/retry** — exponential with jitter; circuit breaker.
9. **Cache layer** — Supabase rows keyed (source, fact, asof).
10. **Test fixtures** — sample responses checked into repo for offline tests.

E.3 Master alpha → data map (which alphas need which providers)

Alpha	Required free	Lifts with paid	Defers
Forensic earnings quality	EDGAR	None	None
Filing red flags	EDGAR	EDGAR full-text plus	None
Dilution / SBC	EDGAR (share counts), yfinance	FMP for SBC explicit	None
Catalyst watch	Finnhub calendar	FMP estimate revisions	Aiera-grade transcripts
Insider clustering	EDGAR Form 4	SecuritiesDB roll-up	None
13F accumulation by quality holders	EDGAR 13F	Aggregator	None
Quality / ROIC	yfinance + EDGAR	FMP key metrics	None
Valuation peers	yfinance / Finnhub + sector mapping	FMP / EOD better fundamentals	None
Macro regime	FRED	None	None
Multi-horizon momentum	yfinance / Polygon	Polygon paid	None
Short interest / options skew	Finnhub limited	Tiingo / Polygon paid	None
ETF overlap	Issuer JSONs / EigenDex	None	None
Earnings transcripts (long form)	Press releases	FMP transcripts (limited)	Aiera (defer)

The system must function on Tier 1 (free) alone; paid tiers are accelerators.

E.4 Reliability & fallback policy

- Every module declares its **required** sources and **optional** sources.
- Each adapter publishes a provider_health row daily.
- A run executes with whatever is available; modules with missing required sources go suppressed=true.
- Decision policy treats suppressed modules as missing, not zero — preserves differentiation.

F. Module-by-module design (full detail)

For each module: purpose, inputs, source requirements, deterministic outputs, data quality gates, suppression behavior, hallucination risks, tests required, how it feeds final action, frontend translation. (Earlier modules condensed; new modules added.)

F.1 data_truth_v1

Purpose: Single contract for every external fact. Every numeric/textual fact carries citation + freshness.

Inputs: Provider adapters (yfinance, finnhub, polygon, FRED, EDGAR, ETF holdings, optional FMP).

Source requirements: Each fact has `source_priority`; conflicts logged.

Deterministic outputs: `TruthRow[]` keyed by `(ticker, fact, asof)`.

Gates: `freshness_band` $\in$ {fresh, stale, expired} from per-fact TTL; expired = suppressed.

Suppression: "provider_outage", "ticker_not_covered", "stale_beyond_ttl", "schema_drift".

Hallucination risks: None — pure deterministic.

Tests: Provider-down simulation, conflict resolution, TTL enforcement, citation completeness, schema-drift alerts.

Feeds: Indirectly via every module.

Frontend: Powers "Evidence" chips and the data-quality banner. Never shown by raw key.

F.2 snapshot_store_v2

Purpose: Append-only per-run snapshots with content-hash addressing. Replaces the implicit snapshot-ness of `agent_runs.allocation`.

Schema: `intel_run_snapshots(run_id, ts, modules_present[], content_hash, schema_version, partial_bool, missing_modules[])` plus per-(run, ticker, module) JSONB rows.

Gates: Reject incomplete snapshots that lack required modules for the run mode.

Suppression: Snapshots may be flagged `partial=true`.

Hallucination risks: None.

Tests: Idempotent writes, hash collisions, partial-snapshot lineage, restore from snapshot, schema-version migration.

Feeds: Every read path.

Frontend: "As of {ts}, {n}/{m} modules ready" chip.

F.3 business_model_v1

Purpose: Plain-English business-model summary; segment / geography mix.

Inputs: 10-K/10-Q business description, segments, geo mix.

Outputs: {segments[{name, revenue_pct}], geo[{region, revenue_pct}], one_line_description, citations}.

Gates: Need ≥ 1 y segment data.

Suppression: Foreign / micro-cap with limited disclosure.

Hallucination risks: Medium — LLM may extrapolate. Mitigation: only summary text from LLM; segments and geo are deterministic from filings.

Tests: Segment sum sanity, geo sum sanity, citation completeness.

Feeds: Source pack; "What this is" drawer line.

Frontend: One sentence at top of drawer.

F.4 return_drivers_v1

Purpose: Decompose expected return into drivers: revenue growth, margin expansion, multiple change, capital return.

Inputs: Multi-year revenue, EPS, FCF, buybacks, dividends, multiples.

Outputs: {driver, contribution_band $\in$ {large, moderate, small, negative}, confidence, inputs_used, inputs_missing} per driver.

Gates: ≥ 2 y history; otherwise suppress per-driver.

Suppression: Per-driver, never global.

Hallucination risks: Naming a "growth" driver on one quarter of data. Gate enforces minimum history.

Tests: Min-history enforcement, contribution sums sanity, driver labels match plain-English.

Feeds: Attractiveness axis.

Frontend: "What could drive returns" — 2–3 bullets.

F.5 quality_moat_v1

Purpose: Standalone quality + moat read.

Inputs: ROIC, FCF margin, gross margin, revenue durability, debt levels, customer concentration if available, moat hints (network effects, switching costs, scale, IP — sourced where possible).

Outputs: {quality_band $\in$ {high, durable, mixed, low}, moat_signals[], reinvestment_runway_band, inputs_used, inputs_missing}.

Gates: ≥ 3 quality inputs; ROIC > WACC proxy for $\geq 3y$ to publish "high" / "durable".

Suppression: ETF / crypto routed to fund_intel_v1.

Hallucination risks: "Compounder" without sustained ROIC.

Tests: No-quality-no-band, ROIC>WACC enforcement, moat signal grounding, ETF routing.

Feeds: Attractiveness axis.

Frontend: "Business quality: {band}; key signs: {...}; what we don't yet measure: {...}".

F.6 management_quality_v1

Purpose: Capital-allocation track record + insider behavior + tenure.

Inputs: CEO/CFO tenure (filings/news), Form 4 buys (signal-significant), buyback timing, M&A accretion, dividend record.

Outputs: {tenure_band, capital_allocation_band, insider_buy_signal, signals[], citations}.

Gates: $\geq 3y$ data.

Suppression: Common for recent IPOs.

Hallucination risks: Narrative on CEO without sourced facts. Mitigation: deterministic facts; LLM may quote.

Tests: Buy-signal calibration, dilution detection, M&A flag.

Feeds: Attractiveness modifier.

Frontend: One-line plain-English summary.

F.7 capital_allocation_quality_v1

Purpose: Numeric scorecard of stewardship.

Inputs: Buybacks at price levels, dividend history, share-count trajectory, M&A history.

Outputs: {capital_alloc_band, signals[], citations}.

Gates: $\geq 3y$ share-count + cash-distribution.

Tests: Dilution detection, accretive vs dilutive M&A, buyback timing band.

Feeds: Quality axis modifier.

Frontend: "Management has {...} over {n}y" sentence.

F.8 earnings_quality_v1 (NEW vs Draft 1)

Purpose: Forensic accounting probes.

Inputs: Multi-quarter income statement, balance sheet, cash flow.

Outputs: {eq_band ∈ {clean, mixed, flagged}, beneish_m, sloan_accruals, cash_conversion_5y, dso_trend, dpo_trend, inventory_days_trend, signals[], citations}.

Gates: ≥4 quarters of complete financials.

Suppression: Financials/REITs/utilities have different patterns; route to industry-specific gates or suppress flag.

Hallucination risks: Calling industries "manipulating" because they legitimately diverge. Mitigation: industry-aware thresholds.

Tests: Beneish-M math, Sloan accruals math, cash-conversion, financial-industry suppression, known fraud-case fixtures.

Feeds: Red-team veto input.

Frontend: "Earnings quality: clean / mixed / flagged" with 1-line why.

F.9 valuation_context_v1

Purpose: Multi-metric valuation read against history AND peers.

Inputs: Trailing/forward PE, P/S, P/FCF, EV/EBITDA, dividend yield, FCF yield, earnings yield (EBIT/EV à la Greenblatt). Baselines: own 5y history, sector median, market median, growth-adjusted, peer-set (constructed by peer_set_v1).

Outputs: {valuation_band ∈ {cheap, fair, full, expensive}, confidence, baselines_used[], why_band, citations}.

Gates: ≥3 metrics OR ≥2 baselines.

Suppression: Suppress band for negative-earnings companies on PE; use FCF/EV instead.

Hallucination risks: "Cheap" without quality context. Mitigation: only used as one axis; never standalone label.

Tests: No single-PE labels, suppression for sparse, sector-median sanity, peer-set traceability, growth-adjusted math.

Feeds: Price/timing axis.

Frontend: "Valuation looks {band} on {n} of {m} measures vs its 5y history; sector context: {...}".

F.10 growth_durability_v1

Purpose: Forward growth assessment with reinvestment-runway lens.

Inputs: Revenue history, segment growth, capex/depreciation, acquisitive vs organic split (if available), TAM proxies.

Outputs: {growth_durability_band, organic_growth_band, runway_band, signals[], citations}.

Gates: $\geq 3y$ revenue + $\geq 2y$ proxies.

Tests: Organic vs acquisitive separation, runway logic.

Feeds: Attractiveness axis.

Frontend: "Growth: {band}; runway: {band}; mostly {organic|acquisitive}".

F.11 margin_trajectory_v1

Purpose: GM/OM/FCF margin trends.

Inputs: $\geq 3y$ income statement.

Outputs: {gm_trend, om_trend, fcfm_trend, peak_trough, citations}.

Gates: $\geq 3y$.

Tests: Trend math, peak/trough detection.

Feeds: Quality axis modifier.

Frontend: Compact "Margins: {direction}" line.

F.12 balance_sheet_resilience_v1

Purpose: Net debt/EBITDA, interest coverage, debt-maturity wall.

Inputs: Most recent balance sheet, debt schedule from filings.

Outputs: {leverage_band, coverage_band, maturity_wall_2y_pct, citations}.

Gates: ≥ 1 quarter detailed BS + debt notes.

Tests: Maturity wall extraction, coverage math.

Feeds: Risk axis input.

Frontend: "Balance sheet: {band}; near-term debt: {...}".

F.13 cashflow_quality_v1

Purpose: OCF vs NI, FCF vs OCF, working capital.

Inputs: ≥ 4 quarters cash flow + income statement.

Outputs: {cfq_band, ocf_to_ni_5y, fcf_to_ocf_5y, wc_trend, citations}.

Gates: ≥ 4 quarters.

Tests: Cash conversion, WC trend math.

Feeds: Earnings-quality / red-team input.

Frontend: "Cash flow quality: {band}".

F.14 dilution_risk_v1

Purpose: Equity issuance, ATM, convertibles, SBC.

Inputs: Share count history, S-1/S-3/424 filings, convertibles, SBC line.

Outputs: {dilution_band $\in$ {clean, ongoing, aggressive}, sbc_pct_fcf_3y, signals[], citations}.

Gates: $\geq 2y$ share-count.

Hallucination risks: Narrative without math. Mitigation: deterministic share-count + SBC%.

Tests: Reverse-split detection, ATM offering flag, convert-bond flag, SBC% math.

Feeds: Red-team veto input; downgrade BUY when band = aggressive.

Frontend: "Dilution: {band}; share count up/down {x}% over {n}y; SBC: {x}% of FCF".

F.15 filing_risk_v1

Purpose: Forensic filing forensics — going-concern, restatements, late filings, auditor changes, RPTs.

Inputs: Recent 10-K/10-Q/8-K, 13D/13G filings, NT-10K/NT-10Q (late filing notifications).

Outputs: {filing_band ∈ {clean, flagged, serious}, flags[], citations}.

Gates: EDGAR coverage; foreign tickers explicitly suppressed.

Hallucination risks: LLM summary of filing it didn't read. Mitigation: deterministic regex/keyword extraction; LLM only quotes.

Tests: Going-concern detector, late-filing flag, auditor-change flag, RPT threshold, restatement detection.

Feeds: Red-team veto input.

Frontend: "Filing risk: clean / flagged / serious" with citation.

F.16 customer_concentration_v1

Purpose: Top customer % of revenue (when disclosed).

Inputs: 10-K segment / risk factors.

Outputs: {top_customer_pct, top3_pct, signals[], citations}.

Gates: Disclosure-dependent.

Suppression: Common — explicit "not disclosed" output.

Tests: Disclosure parsing, threshold flags.

Feeds: Risk axis modifier.

Frontend: "Customer concentration: {...}" if disclosed.

F.17 regulatory_litigation_risk_v1

Purpose: Material litigation, regulatory actions.

Inputs: 10-K legal proceedings, 8-K, news (filtered).

Outputs: {reg_lit_band, active_cases[], citations}.

Gates: EDGAR + news.

Tests: Material vs immaterial classification; date freshness.

Feeds: Risk axis.

Frontend: "Active litigation: {none / list}".

F.18 geopolitical_supply_chain_v1

Purpose: Geographic mix, sanction exposure.

Inputs: 10-K geo segments, news.

Outputs: {geo_mix, sensitive_exposure_pct, signals[]}.

Gates: Disclosure-dependent.

Feeds: Risk axis modifier.

Frontend: Optional drawer line.

F.19 catalyst_watch_v1

Purpose: Forward-looking scheduled catalysts + probabilistic events.

Inputs: Earnings calendar (Finnhub), dividend calendar, optional FDA/regulatory event feeds, optional macro events (FOMC, CPI).

Outputs: {catalysts[{type, expected_date, importance_band, confidence, citations}]}

Gates: Calendar present.

Hallucination risks: LLM inventing catalysts. Mitigation: LLM may only reference catalysts present in source pack.

Tests: Date accuracy, no fabricated catalysts.

Feeds: Timing modifier; alerting.

Frontend: "Next 90 days: earnings {date}, ex-div {date}, regulatory {window}".

F.20 technical_confirmation_v1

Purpose: Multi-horizon momentum + drawdown + RSI as *confirmation* of fundamental thesis.

Inputs: Price history.

Outputs: {trend_horizons: {20d, 60d, 125d, 250d -> band}, rsi_14, drawdown_from_52wh, rs_vs_spy_band, divergence_flag, confidence}.

Gates: ≥ 6 mo history.

Hallucination risks: LLM extrapolating chart patterns. Mitigation: deterministic; LLM only quotes.

Tests: Trend classification reproducibility, RS math, divergence logic.

Feeds: Tie-breaker. Strong-down + bullish fundamentals = HOLD not BUY.

Frontend: One sentence + chip per horizon.

F.21 sentiment_noise_filter_v1

Purpose: News tone, source-quality-weighted; social demoted.

Inputs: Finnhub news + (optional) curated source list. NO Reddit/Twitter as core signal.

Outputs: {sentiment_band, source_count, dispersion, top_articles_with_citations}.

Gates: Min source count.

Suppression: Default suppressed at low source count.

Hallucination risks: Treating sentiment as fact. Mitigation: tertiary signal only.

Tests: Min-source enforcement, no-social contamination, source-quality weighting.

Feeds: Tertiary modifier.

Frontend: "Recent news tone" if surfaced; consider hiding by default.

F.22 short_options_tape_v1 (NEW vs Draft 1)

Purpose: Short interest + implied vol skew as asymmetric tape signals.

Inputs: Finnhub or Polygon short interest; yfinance options chain (free, OK quality on majors).

Outputs: {short_interest_pct, days_to_cover, iv_skew_band, citations}.

Gates: Provider-dependent; suppress for illiquid options or thin short interest.

Tests: Skew math, days-to-cover, suppression on illiquid.

Feeds: Red-team modifier (high short + earnings catalyst = enhanced bear case).

Frontend: Optional drawer chip.

F.23 factor_exposure_v1 (NEW vs Draft 1)

Purpose: Per-name and portfolio factor loadings (value, quality, momentum, size, low-vol, growth).

Inputs: Factor returns proxies (free indexes), holdings, ETF passthrough.

Outputs: Per ticker {factor_loadings: {value, quality, momentum, size, low_vol}, primary_factor}.
Portfolio aggregate {factor_mix}.

Gates: Universe-aware.

Suppression: Crypto / atypical assets.

Tests: Factor decomposition sanity; portfolio aggregation arithmetic.

Feeds: Portfolio governor; "fit" signal; "would this trade increase concentration?" check.

Frontend: Portfolio-level chip in Command Center.

F.24 fund_intel_v1

Purpose: Asset-class router: ETF/fund role, holdings, overlap, expense, replication; for crypto: market cap, supply schedule, exchange dispersion.

Inputs: Issuer holdings JSONs, ticker registry, CoinGecko.

Outputs: {asset_class, fund_role ∈ {core_index, factor_tilt, sector, theme, income, speculative}, expense_ratio, aum, replication, overlap_with_portfolio[]}.

Gates: Holdings file or recognized ticker.

Tests: ETF role classification, overlap-pct math, crypto routing.

Feeds: Quality / risk axes for funds; portfolio overlap.

Frontend: "ETF role: {...}; overlaps with: {tickers}; expense ratio: {x}".

F.25 peer_set_v1 (NEW)

Purpose: Constructed peer set per ticker for relative valuation.

Inputs: Sector + sub-industry mapping; size band; quality band; country.

Outputs: {peer_tickers[], construction_rule, conflicts}.

Gates: Need ≥ 3 valid peers to publish.

Hallucination risks: Bad peer sets. Mitigation: deterministic construction rules; surface peers in drawer for user to inspect.

Tests: Peer-set determinism, size-band rules.

Feeds: valuation_context_v1.

Frontend: "Peers used: {tickers}" in drawer.

F.26 insider_inst_v1 (NEW)

Purpose: Form 4 insider buy clusters; 13F changes from a curated list of "respected holders" (configurable).

Inputs: EDGAR Form 4 (parsed); 13F holdings (parsed).

Outputs: {insider_cluster_band, recent_signal_buys[], top_inst_changes[], citations}.

Gates: EDGAR coverage.

Hallucination risks: "Famous investor X bought" — must not appear as primary UI. Mitigation: surface as evidence in drawer, never as cosplay.

Tests: Cluster detection (3+ insiders within 30 days, etc.); 13F delta math.

Feeds: Modifier on attractiveness; opportunity radar input.

Frontend: Drawer evidence chip; not a card-level headline.

F.27 macro_regime_v1

Purpose: Coarse regime read.

Inputs: FRED 10Y, 2Y, fed funds, U-rate, ISM proxies, USD, HY spread proxy.

Outputs: {regime $\in$ {expansion, mid, late, contraction, uncertain}, asset_class_tilts (informational), confidence}.

Gates: ≥ 3 series fresh.

Tests: Regime change detection on known dates.

Feeds: Modifier on conviction; "why now / why not now".

Frontend: Once at top of Portfolio Command Center.

F.28 risk_monitor_v1 (NEW)

Purpose: Stress test the current portfolio and flag concentration.

Inputs: Holdings, factor exposures, sector weights, historical returns.

Outputs: {stress_2008_pl_band, stress_2020_pl_band, stress_2022_pl_band, +rate_shock_pl, +usd_shock_pl, +oil_shock_pl, hidden_concentration_flags[]}.
+rate_shock_pl, +usd_shock_pl, +oil_shock_pl, hidden_concentration_flags[]}.

Gates: Need ≥3y price history on majority of holdings.

Suppression: Crypto excluded from equity stress windows.

Tests: Replay accuracy on known periods, concentration detection.

Feeds: Conviction modifier; portfolio-level alerting.

Frontend: Portfolio Command Center "stress" chip; drawer expansion.

F.29 behavioral_guardrails_v1 (NEW)

Purpose: Counteract overconfidence, anchoring, loss aversion, disposition effect.

Inputs: User decision history, current actions, holding periods.

Outputs: {flags: [{type ∈ {repeated_addition_to_loser, sell_winner_no_thesis_change, anchoring_on_cost, recency_overreaction}, severity, message, citations}]}.
anchoring_on_cost, recency_overreaction}, severity, message, citations}]}

Gates: Decision history present.

Tests: Each bias type triggers on canonical fixtures.

Feeds: Soft warnings before user clicks "Use AI plan"; second-look prompts.

Frontend: "Heads up" prompts; never blocks unless user opted in.

F.30 thesis_tracker_v1 (NEW)

Purpose: Per-position thesis with must_be_true[], would_change_view[], drift detector.

Inputs: User-written thesis (or LLM-drafted, user-approved); fundamental snapshots.

Outputs: {thesis_status ∈ {intact, drifting, broken}, drift_signals[], last_review_ts}.
drift_signals[], last_review_ts}

Gates: Thesis must exist; we provide draft if missing.

Tests: Drift detection on canonical examples; broken-thesis detection.

Feeds: Decision policy (broken → SELL eligibility); What-Changed.

Frontend: "Thesis check" pill; "drift detected" delta.

F.31 delta_engine_v1 **(NEW)**

Purpose: Compute snapshot diffs and surface what changed.

Inputs: Two snapshots (latest + previous).

Outputs: {action_changes[], red_flags_added[], red_flags_cleared[], conviction_changes[], thesis_drifts[], radar_promotions[], radar_demotions[], data_quality_changes[]}.

Tests: Diff completeness, dedupe logic.

Feeds: What-Changed UI; alerting.

F.32 committee_v1 **(LLM)**

Detailed in Section H.

F.33 decision_policy_v1

Detailed in Section I.

F.34 opportunity_radar_v1

Detailed in Section G.

F.35 portfolio_governor_v1

Detailed earlier; Layer 3.

F.36 feedback_v1 **(NEW — Layer 8)**

Purpose: Track post-decision performance; compute decision-quality metrics; surface in UI.

Inputs: Decision logs (existing tables decision_logs*), prices, benchmarks, snapshots.

Outputs: Per ticker per decision {horizon_returns: {7d, 30d, 90d, 180d, 365d}, vs_benchmark, vs_alternative_held, decision_quality_band ∈ {good, neutral, bad}, attribution: {alpha_factor, sector, idiosyncratic}}. Portfolio aggregate {brinson: {allocation_effect, selection_effect}, factor_attribution}.

Gates: Decision history present.

Tests: Brinson math; horizon returns; benchmark alignment.

Feeds: Calibration of conviction (system learns it tends to over-conviction in regime X); user trust UI.

Frontend: Drawer "How was the system on this name" + Phase-2 Performance tab.

F.37 replay_v1 (NEW — Layer 8)

Purpose: Run historical points-in-time through the engine for backtest-like validation.

Inputs: Historical snapshots; archived data.

Outputs: {replay_runs[], replay_diffs[]}.

Gates: Historical data depth.

Hallucination risks: Look-ahead bias. Mitigation: every fact in a replay run must have ts_published <= replay_asof.

Tests: Look-ahead detection; replay determinism.

Frontend: Optional admin/Settings tab.

G. Opportunity Radar design (full detail)

The Radar is what separates a position monitor from an investment committee. Its job is to surface high-quality, source-backed candidates outside the held universe — without ticker spam and without LLM hallucination.

G.1 Universe definition (universe_v1)

A closed, expansion-controlled universe. LLM cannot suggest tickers outside it.

Layer	Source	Approx size	Refresh
L1 Held	User holdings	~34	Realtime
L2 Pass-through	ETF holdings of held ETFs	~1500	Monthly
L3 Watchlist	Manual additions	dozens	On-edit

Layer	Source	Approx size	Refresh
L4 Sector benchmarks	Top N by market cap of GICS sectors	~500	Monthly
L5 Quality compounder seed list	Curated by ROIC>WACC heuristic; refreshed quarterly	200	Quarterly
L6 Macro/theme tilts	Names tagged to current regime tilts	varies	Quarterly
L7 Index baselines	S&P 500 + S&P 600 + Russell 1000 constituents	~2000	Monthly
L8 Activist / 13F-quality holders' books	Curated 13F roll-up, recent additions	~500	Quarterly

Total deduped universe target: ~3000 tickers. The radar processes a much smaller candidate set after deterministic gating.

G.2 Screens (parallel; merged and ranked)

Each screen is a deterministic filter. Each emits RadarCandidate rows with {ticker, screen_id, score_band, signals[], inputs_used, suppression_reason | null}.

- Quality compounder screen.** ROIC > X for 5y AND FCF margin > Y AND share count flat/down 3y AND earnings_quality_v1 band ≥ "mixed" AND filing_risk_v1 = "clean". Output: ranked by sustainability and reinvestment runway.
- GARP screen.** Forward growth ≥ X% AND PEG ≤ Y AND quality ≥ "durable" AND filing clean.
- Magic-formula adapted screen** (Greenblatt). Earnings yield (EBIT/EV) decile + ROC decile composite, top-N. Filtered by filing_risk = clean and earnings_quality ≥ "mixed".
- Valuation dislocation screen.** Valuation = "cheap" AND quality ≠ "low" AND filing clean AND no aggressive dilution. Cheap-and-good, not cheap-and-garbage.
- Catalyst screen.** Near-term scheduled catalyst (earnings, ex-div, regulatory) AND fundamentals not deteriorating AND filing clean.
- Insider clustering screen.** ≥3 distinct insider buys ≥ significance threshold within 60 days AND filing clean. Signal: aligned-incentive buying.
- Smart money 13F screen.** New positions or material additions across multiple curated quality holders AND filing clean.
- Estimate revisions screen** (paid tier or fallback). Upward revisions to forward EPS/revenue + price not yet reflecting.
- Momentum-with-fundamental-confirmation screen.** Strong RS vs SPY 6m/12m AND quality ≥ "mixed" AND no critical red-team. Filters meme momentum.
- ETF upgrade screen.** For each held ETF, find better-fit ETFs (lower cost, less overlap, better factor balance vs IPS).
- Sector/theme leader screen.** Top quality+momentum names per under-represented sector vs IPS targets.
- Hidden gem screen.** Smaller market cap (>\$300m, <\$5b) + clean filings + 3y ROIC > cost-of-capital + low coverage. Strict evidence gating; suppressed by default if any axis thin.

G.3 Avoid-list constraints (rejection rules)

- Aggressive dilution band → reject.
- Going-concern flag → reject.
- Filing flags severe → reject.
- Recent reverse split + EPS < 0 → reject.
- Earnings quality "flagged" → reject unless screen-specific override.
- Customer concentration > X (if known) → flag, not auto-reject.
- Pump signal: high social mentions + thin fundamentals → reject.
- Asset-class = "speculative" beyond user policy → reject.
- Foreign tickers without EDGAR coverage → demote (require manual whitelist).

G.4 Ranking (deterministic, then committee-gated)

Two-pass ranking. Never a single weighted score.

Pass 1 — Deterministic Pareto + tiered rules:

1. For each candidate, compute axis bands: EQ, AT, PR, FIT, technical_band, factor_fit_to_portfolio_band.
2. Pareto rank: a candidate dominates another only if it is at least as good on every axis and strictly better on one.
3. Filter to "non-dominated front" + "near-front" (within 1 ordinal of front on at most 1 axis).
4. Tier into: **Tier A** = strong (≥3 axes "ok+"), **Tier B** = mixed, **Tier C** = thin (no LLM call).
5. Per-screen quota: max N per screen to avoid one screen dominating.

Pass 2 — LLM committee gating (Tier A + select Tier B):

- Top 15 candidates after Pass 1 → committee express mode (Bull+Bear synthesized).
- Top 5 → committee standard mode.

Final ranking:

- Sort by tier first, then by axis-coverage strength, then by fit_improvement_band (does this improve portfolio factor balance?).

G.5 Suppression cascades

- If data_truth_v1 is degraded for >50% of universe → radar runs with banner "Data thin; results limited".
- If LLM committee unavailable → ranking still happens; cards show "Committee deferred" badge.
- If a screen has zero candidates after gates → empty state with screen name and reason.

G.6 Cadence and cost controls

- **Full radar pass:** weekly (cron).
- **On-demand single-ticker:** any time (drawer "Analyze this name" button — Phase 2).
- **Universe-wide LLM call:** only on top 15 after deterministic ranking.
- **Cache committee output by source-pack content hash** — re-runs use cache if pack unchanged.
- **Per-run token cap.** Cap configurable; hit-cap triggers graceful abort with what-completed-so-far snapshot.

G.7 Hallucination control

- Closed universe — out-of-universe ticker = validator rejection.
- LLM bull/bear must cite source-pack `fact_ids` for every claim.
- Output schema requires `screen_id` reference — system can trace which screen surfaced each candidate.
- Manual review hook: user can flag a radar suggestion as "implausible"; reflagging bumps a per-screen confidence prior down.

G.8 UI output

- "New opportunities" section, max 5 default ("show more" expands to 15).
 - Each row: ticker + name, action badge (WATCH/BUY), one-sentence why, one-sentence risk, evidence chip, "from screen X" tag.
 - Drawer: full Bull/Bear, citations, fit explanation, "would this improve portfolio?" answer.
 - Empty state: "No new opportunities cleared the bar this run."
-

H. LLM committee design (full detail)

The committee is the synthesis layer that turns a deterministic source-pack into a debate the user can read. It never produces the action label, never invents numbers, and is always validator-gated.

H.1 Personas

1. **Bull Analyst.** Strongest case for ownership. Allowed to use only source-pack facts and module outputs. Required to cite `fact_id` or `module_id` per key point.
2. **Bear Analyst.** Strongest case against ownership. Different temperature than bull; ideally different model in `full` mode. Required to be non-empty; "no risks" rejected.
3. **Forensic Accountant.** Reads `earnings_quality_v1`, `filing_risk_v1`, `dilution_risk_v1` outputs and explains in plain English. Skipped when those modules return suppressed.
4. **Valuation Analyst.** Reads `valuation_context_v1` and `peer_set_v1`. Translates band to plain English; explains drivers.

5. **Macro/Regime Analyst.** Reads `macro_regime_v1`. Used per-portfolio (one summary), not per-ticker.
6. **Technical/Momentum Analyst.** Reads `technical_confirmation_v1`. Single sentence; never primary driver.
7. **Risk Officer.** Active veto search. Allowed to flag `critical` only when sourced.
8. **Portfolio Manager (PM).** Synthesizes. Produces one paragraph manager note + `confidence_band` + `what_would_change_the_view` + `why_now` + `why_not_now`. Cannot produce action.
9. **CIO / Judge (in full mode only).** Adjudicates explicit dissent; renders a final dissent statement when consensus is not reached. **Never overrides the deterministic decision policy**; just shapes the narrative.

H.2 Allowed / forbidden claims

Allowed (per persona):

- Quote any source-pack fact by `fact_id`.
- Cite any module output by `module_id`.
- Synthesize, summarize, and contextualize.
- Express qualitative judgment ("management appears disciplined based on `capital_allocation_quality_v1` band 'durable'").
- Flag uncertainty and ask for more data.

Forbidden (validator rejects):

- Any number not present in source pack.
- Any claim without `fact_id` or `module_id` reference.
- Forbidden phrases under thin EQ: "buy", "accumulate", "high-conviction idea", "strong buy", "load up".
- Raw metric keys (`fcf_margin`, `roic_ttm`, `ev_ebitda`).
- Precision price targets ("185"). Allowed: scenarios with stated assumptions, labeled "scenario, not target".
- Famous-investor cosplay ("Buffett would buy this"). Allowed: "13F filings show holder X added to this position" with citation.
- Out-of-universe ticker references in radar.
- Pretending data exists when it does not.
- Imperatives that imply action ("you should buy"). Use neutral language ("the case for ownership rests on ...").

H.3 Source pack format (typed schema)

```
{
  "schema_version": "1.0",
  "ticker": "AAPL",
  "asof": "2026-05-04T18:00Z",
  "asset_class": "equity",
  "policy_version": "v1.3",
  "user_context": {
    "current_weight": 0.072,
    "tax_lots_summary": null,
    "last_decision": {"action": "HOLD", "ts": "2026-04-27"},
    "thesis_status": "intact"
  },
  "facts": [
```

```

{"id": "f1", "name": "trailing_pe", "value": 28.4, "unit": "x", "source": "yfinance",
 "ts_published": "2026-05-04T17:00Z", "url": "...", "confidence": 0.9, "freshness": "fresh"},
...
],
"modules": {
  "quality_moat_v1": {
    "module_id": "m_qm",
    "band": "durable",
    "signals": [{"label": "ROIC > WACC for 7y", "citations": ["f4", "f5"]}],
    "inputs_used": ["roic_5y", "wacc_proxy", "fcf_margin_5y"],
    "inputs_missing": [],
    "data_quality": {"coverage_pct": 0.92, "freshness_band": "fresh", "confidence": "high"},
    "suppressed": false
  },
  "valuation_context_v1": { ... },
  "earnings_quality_v1": { ... },
  "filing_risk_v1": { ... },
  "dilution_risk_v1": { ... },
  "return_drivers_v1": { ... },
  "technical_confirmation_v1": { ... },
  "catalyst_watch_v1": { ... },
  "factor_exposure_v1": { ... },
  "portfolio_fit_v1": { ... },
  "macro_regime_v1": { ... }
},
"deltas_since_last_run": [ ... ],
"screens_surfaced": ["quality_compounder"]
}

```

H.4 Output schema (typed; validator-enforced)

```

{
  "schema_version": "1.0",
  "ticker": "AAPL",
  "personas": {
    "bull": {
      "thesis": "...",
      "key_points": [
        {"text": "...", "citations": ["f4", "m_qm"]}
      ],
      "confidence": "MEDIUM"
    },
    "bear": {
      "thesis": "...",
      "key_points": [
        {"text": "...", "citations": ["f12", "m_eq"]}
      ],
      "confidence": "MEDIUM"
    },
    "forensic": {"summary": "...", "flags": [], "citations": ["m_eq", "m_fr"]},
    "valuation": {"summary": "...", "band_explanation": "...", "citations": ["m_vc"]},
    "technical": {"summary": "...", "citations": ["m_tc"]},
    "risk_officer": {"critical_flags": [], "non_critical_flags": [], "citations": [...]},
    "pm": {
      "manager_note": "...",
      "confidence": "MEDIUM",
      "why_now": "...",
      "why_not_now": null,
      "what_would_change_the_view": ["...", "...", "..."]
    },
    "cio": {"adjudication": null, "dissent": null}
  },
  "tokens_used": 4521,
  "cost_estimate_usd": 0.018,
  "validator_passes": ["citation_resolution", "numeric_grounding", "forbidden_phrases"]
}

```


H.5 Tiered modes and cost discipline

Mode	Personas called	Use case	Approx cost (Sonnet/Opus mix)
lite	Bull+Bear synthesized in 1 call + PM in 1 call	Daily refresh on stable holdings	~\$0.005/ticker
express	Bull, Bear, PM	Routine refresh	~\$0.01/ticker
standard	Bull, Bear, Risk, Forensic, Valuation, PM	Default for full Run Agents on holdings	~\$0.03/ticker
full	All 9 incl. CIO	Special cases: thesis review, large position changes, radar Tier A	~\$0.06/ticker

Cost cap algorithm:

1. At run start, total token budget = tier_budget × n_tickers.
2. Each persona call deducts from a per-run pool.
3. If pool < threshold, mode downgrades automatically (full → standard → express).
4. If pool exhausted, remaining tickers marked "committee deferred"; deterministic decision still produced.

H.6 Validators

- **Citation resolution:** Every citation ID must exist in source pack.
- **Numeric grounding:** Any number in text must match a source-pack value within tolerance.
- **Forbidden phrases:** List enforced under thin EQ.
- **No raw keys:** Regex against known raw metric names.
- **No price targets:** Regex against $\$d+(\backslash.\backslashd+)?$ followed by "target" / "fair value" without "scenario".
- **Bear non-empty:** Bear case must have ≥1 key point or get rejected and retried.
- **Schema conformance:** JSON-schema validation.
- **Out-of-universe rejection:** Radar suggestions cross-checked against universe_v1.
- **Red-team validator:** Independent post-pass that re-runs red_team_v1 rubric over the output and amends if missed flags found.

H.7 Disagreement handling

- **Soft disagreement** (Bull + Bear coherent, different conclusions): preserve both views in drawer; PM synthesizes; user sees the disagreement.
- **Hard disagreement** (Risk Officer flags critical sourced veto): action capped at TRIM (or SELL if filing/dilution criteria met) regardless of bull persona.
- **Module-vs-LLM disagreement** (e.g., LLM Valuation persona disagrees with valuation_context_v1 band): LLM may surface its disagreement with reasoning, but the band remains the deterministic one. The decision policy uses the deterministic band.

H.8 When LLM is skipped

- Source pack incomplete beyond threshold → committee skipped; deterministic decision still produced.
 - Per-run cost cap exhausted → remaining tickers skipped.
 - Cache hit on identical source-pack content hash → no LLM call; cached output served.
-

I. Decision policy (full state machine)

The policy is a deterministic table over four primary axes plus modifiers and vetoes. It is the only producer of action.

I.1 Axes

```
EQ   = evidence_quality   ∈ {thin, ok, strong}
AT   = attractiveness     ∈ {weak, ok, strong}
PR   = price_context       ∈ {expensive, full, fair, cheap, suppressed}
FIT  = portfolio_fit       ∈ {blocked, underweight, on_target, overweight, breach}
THESIS = thesis_status     ∈ {intact, drifting, broken, missing}
```

Modifiers:

```
TECH ∈ {confirms, neutral, opposes}
CAT  ∈ {none, near_positive, near_negative}
REG  ∈ {risk-on, mid, risk-off}
EQT  ∈ {neutral, low_volatility, high_volatility}
```

Vetoes:

```
RT   = red_team_severity ∈ {none, low, medium, high, critical}
EQ_RED = earnings_quality_band ∈ {clean, mixed, flagged}
FILING_RED = filing_band ∈ {clean, flagged, serious}
DIL_RED = dilution_band ∈ {clean, ongoing, aggressive}
```

I.2 Held-ticker action rules (priority order)

1. **SELL** triggers if any: - THESIS = broken (sourced) AND user-policy allows SELL. - FILING_RED = serious AND RT = critical AND citations present. - DIL_RED = aggressive AND THESIS ≠ intact. - EQ_RED = flagged AND RT ≥ high AND citations present. - User policy override.

1. **TRIM** triggers if any (and SELL not triggered): - FIT ∈ {overweight, breach}. - RT ≥ high (sourced) without SELL trigger. - PR = expensive AND AT ≠ strong. - THESIS = drifting AND PR ∈ {full, expensive}. - CAT = near_negative AND TECH = opposes AND AT ≠ strong.

1. **BUY** triggers if all: - EQ ≥ ok. - AT ≥ ok. - PR ∈ {fair, cheap} (or PR = suppressed AND AT = strong AND EQ = strong — capped MEDIUM). - RT ≤ medium. - FIT ≠ blocked AND FIT ∈ {underweight, on_target} (BUY at on_target only if MEDIUM cap). - TECH ≠ opposes (or TECH = opposes downgrades to HOLD with "wait for confirmation"). - THESIS ∈ {intact, missing} (missing draws a draft thesis flag in UI).

1. **HOLD** is the default. Never an apology.

1. **Conviction band:** - HIGH: EQ = strong AND AT = strong AND ≥ 3 trusted axes AND RT $\leq$ low AND FIT $\in$ {underweight, on_target} AND TECH = confirms. - MEDIUM: BUY criteria met, but at least one axis ok rather than strong, or TECH neutral. - LOW: BUY criteria barely met, or PR suppressed, or new IPO, or volatility high.

I.3 Non-held (radar) action rules

1. **AVOID** triggers only if sourced disqualifying flag (going-concern, restatement, aggressive dilution, earnings quality flagged with citation, customer-concentration extreme).
2. **WATCH** triggers if EQ $\geq$ ok AND AT $\geq$ ok AND not AVOID. Default for surfaced radar candidates that don't meet BUY bar.
3. **BUY** (radar) triggers only if held-BUY criteria met standalone AND FIT $\neq$ blocked AND policy allows new position AND there is a clear capital priority (better than worst-fit existing holding).

I.4 Hard suppression rules

- EQ = thin $\rightarrow$ BUY forbidden. HOLD allowed; conviction = LOW max.
- PR = suppressed $\rightarrow$ BUY at MEDIUM max.
- TECH = strong-down AND AT = ok $\rightarrow$ BUY downgraded to HOLD with "wait for confirmation".
- New IPO / <6mo history $\rightarrow$ BUY at MEDIUM max.
- Asset-class = crypto/speculative $\rightarrow$ action capped at HOLD unless explicit user override and within speculative cap.
- EQT = high_volatility $\rightarrow$ conviction capped MEDIUM regardless of other axes.
- REG = risk-off AND AT $\neq$ strong $\rightarrow$ conviction capped MEDIUM.

I.5 Required outputs (always emitted)

```
DecisionV3 = {
  ticker: ...,
  action: BUY | HOLD | TRIM | SELL | WATCH | AVOID,
  conviction_band: HIGH | MEDIUM | LOW,
  why_now: str | null,
  why_not_now: str | null,
  what_would_change_the_view: [str, str, str],
  blockers: [str],             // axis blockers
  vetoes: [str],               // sourced vetoes
  suppression_reasons: [str],  // missing-data
  source_pack_hash: str,
  policy_version: str,
  rationale: str               // plain English summary
}
```

I.6 Behavioral guardrails layer (post-policy)

After the policy emits a decision, behavioral_guardrails_v1 may *flag* (not change) the decision:

- BUY-more on a position that already lost 15%+ from cost without thesis change $\rightarrow$ "Second look" flag.

- TRIM/SELL on a winner without fundamental deterioration → "Second look" flag.
- High-frequency reversal (BUY this week, SELL next week, BUY again) → "You've reversed this position 3 times in 30 days" flag.

These flags do not change action; they prompt the user to confirm with reasoning before Deploy executes.

I.7 The HOLD-34 canary (regression rule)

A run is rejected if `count(action == HOLD AND conviction == LOW) == count(active_tickers)` for a snapshot with at least one published thesis dimension. CI fails. The decision policy must emit at least 2 distinct actions OR at least 2 distinct convictions on any non-degenerate snapshot.

J. Frontend product contract (full detail)

A clean, beautiful, beginner-friendly UI that hides complexity. The opposite of Bloomberg. The opposite of a giant report. The opposite of broker noise.

J.1 Core principles

1. **Plain English everywhere.** No raw metric keys. No advanced jargon. Reading age 14.
2. **One canonical action.** BUY/HOLD/TRIM/SELL for held; WATCH/BUY/AVOID for non-held. Nothing else surfaces.
3. **Citations by default.** Hover or click to see the source for any number.
4. **Suppression visible.** "We don't have this yet" is shown, not silently hidden.
5. **Confidence in dots, not percentages.** 1–3 dots map to LOW/MEDIUM/HIGH.
6. **No precision price targets.** Scenarios with assumptions OK, labeled as such.
7. **Cost basis lives in drawer, not on card.** Anti-anchoring.
8. **Bull and Bear shown side by side.** Anti-confirmation-bias.
9. **What changed gets a first-class section.** Anti-recency-bias by also showing 6m view.
10. **Empty states are affirmative.** "No new high-conviction action this run" is good news, not a failure.

J.2 Page structure (recommendations route)

```
[ Header: Portfolio Command Center ]
[ Section A: Best Buys (carousel, 0-5) ]
[ Section B: Trim/Sell Desk (0-5) ]
[ Section C: Current Holdings (filter tabs ALL/BUY/HOLD/TRIM/SELL) ]
[ Section D: Opportunity Radar (0-5 default; expand to 15) ]
[ Section E: What Changed Since Last Run (expandable) ]
[ Footer: Data Quality Banner + Snapshot ts ]
[ Drawer: Ticker Detail (slides from card click) ]
```

J.3 Portfolio Command Center

- **Regime line:** "Late-cycle, rates plateauing. Bias to quality and resilient cash flows." (one sentence, sourced).
- **Capital priority chips:** counts for `urgent_buy`, `opportunistic_buy`, `trim_first`, `sell_first`. Click → filtered list.
- **Stress chip:** "If 2008 repeats: portfolio down ~31% (vs S&P ~37%). Click to expand."
- **Factor mix chip:** "Tilt: quality-momentum. Concentration risk: tech-megacap 42%."
- **Data quality banner:** "{n} of {m} modules ready as of {ts}."
- **Run controls:** "Refresh now" tier selector (Quick / Standard / Full); cost estimate next to button.

J.4 Best Buys (Section A)

- 0–5 cards. Sourced from held BUY (≥MEDIUM conviction) + radar BUY (cleared standalone bar AND FIT allows).
- **Card fields:** ticker + name, action badge (BUY), conviction dots, one-sentence WHY, one-sentence RISK, fit chip ("underweight by 1.5%"), evidence chip ("strong coverage"), "from radar" tag if non-held.
- **Empty state:** "No new high-conviction Buys this run. The portfolio is on track. Check Holdings for ongoing Buys at lower conviction."

J.5 Trim/Sell Desk (Section B)

- 0–5 cards. Sourced from held TRIM/SELL.
- **Card fields:** ticker, action badge (TRIM/SELL), reason code chip ("at concentration cap" / "thesis broken: sourced filing flag" / "valuation expensive, attractiveness only ok"), one-sentence RISK of holding, behavioral guardrail flag if any.
- **Empty state:** "No trims or sells this run."

J.6 Current Holdings (Section C)

- **Filter tabs:** ALL / BUY / HOLD / TRIM / SELL with counts. Locked. Posture buckets forbidden by regression test.
- **Card fields per ticker:**
 - Action badge.
 - Conviction dots.
 - One-sentence WHY (PM note).
 - One-sentence RISK (Risk Officer flag).
 - One-line "what would change the view" (top of three).
 - Fit chip.
 - Evidence chip.
- Click → Ticker Detail Drawer.

J.7 Opportunity Radar (Section D)

- Up to 5 rows by default; "Show 10 more".
- Each row: ticker + name, action badge (WATCH or BUY only), conviction dots, "from screen X" tag, one-sentence why, one-sentence risk, evidence chip, fit-improvement chip.
- AVOID rows hidden by default; "Show avoided" toggle reveals them with sourced reason.
- Empty state: "No new opportunities cleared the bar this run. Universe scanned: {n} names, {k} screened, {m} surfaced for review, all rejected for {reason summary}."

J.8 Ticker Detail Drawer

Sections in order:

1. **Snapshot.** Ticker, name, asset class, role in portfolio, current weight vs target, last refresh, snapshot id.
2. **What we think (PM note).** One paragraph. Action + conviction.
3. **Why** (Bull case). Bullet list with citations.
4. **What could go wrong** (Bear case + Risk Officer). Bullet list with citations.
5. **What would change the view.** 2–4 concrete conditions.
6. **Why now / Why not now.** One sentence each (or only one if applicable).
7. **How it fits my portfolio.** Plain English fit explanation.
8. **Evidence check.** Reliable / Missing chips with plain-English labels.
9. **Catalysts ahead.** Earnings/ex-div/regulatory.
10. **What changed since last run.** Delta diff for this ticker.
11. **Thesis check.** `must_be_true[]`, `would_change_view[]`, drift status.
12. **Performance feedback** (Phase 2). "How was the system on this name?" with horizon returns vs benchmark.
13. **Cost basis & lots** (collapsed by default; anti-anchoring).
14. **Forensic accountant view** (collapsed). Earnings quality, dilution, filing risk in plain English.

No raw metric keys anywhere. No business read. No score numbers. No precision price targets.

J.9 What Changed Since Last Run (Section E)

- **Promotions:** tickers whose action upgraded (HOLD→BUY, TRIM→HOLD, etc.).
- **Demotions:** tickers whose action downgraded.
- **New radar entries.** With "from screen X" tag.
- **Cleared radar entries.** With reason.
- **New red flags** (sourced).
- **Cleared red flags.**
- **Thesis drift detected.**
- **Data quality changes** (e.g., "Filing risk module came back online for 12 names").

J.10 Empty / loading / error states

- **Empty:** Always affirmative. Never "no results" by itself.
- **Loading:** Skeleton cards with last-refresh chip; never blank.
- **Error:** "We couldn't refresh {module}. Last good data: {ts}. Run continues with reduced coverage." Never silent failure.

J.11 Mobile behavior

- Stack sections; carousel becomes vertical list.
- Drawer becomes full-screen modal.
- Filter tabs become a single dropdown.

J.12 Trust preservation

- Show "as of" timestamp on every card.
- Show provider next to any number on hover.
- Show "this is based on free public data; coverage is limited" disclosure in Settings.
- Show "this is not financial advice" disclaimer once per session in footer.
- Never auto-execute; never suggest user execute without reading the bear case.
- Always offer "Why was this card hidden?" when filters hide cards.

J.13 Visual rules

- BUY = green dot + word "Buy". TRIM = amber. SELL = red. HOLD = neutral. WATCH = blue (Radar only). AVOID = grey (Radar only).
- Conviction = 1–3 dots, never percentage.
- Evidence chip: filled (strong) / outlined (ok) / dashed (thin).
- Suppression: explicit "we don't have ..." line; never silent.
- No "make a recommendation" mega-button. Only "Refresh now".

J.14 What the UI must NOT show

- Raw metric keys.
- Business Read wall.
- Posture buckets in filter tabs.
- "Famous investor X bought this" as primary headline.
- Real-time tickers ticker tape.

- Social media feed.
 - Precision price targets.
 - Confidence as percentage.
 - Backend module names ("from quality_moat_v1").
 - Per-ticker score numbers.
-

K. Feedback / backtesting / validation system

Without measurement, the system cannot improve. Without replay, the system cannot be validated. This is Layer 8.

K.1 Decision logs

Every decision the policy emits is logged with:

- `decision_id`, `ticker`, `action`, `conviction`, `policy_version`, `source_pack_hash`, `snapshot_id`, `ts`.
- Axes values used, vetoes, blockers, suppressions, behavioral flags.
- LLM committee output (if any) by hash.
- User action: accepted / overridden (with reason) / ignored.
- Deploy handoff payload emitted.

K.2 Post-decision performance windows

For each decision: 7d, 30d, 90d, 180d, 365d returns vs:

- Buy-and-hold the alternative (if HOLD vs BUY).
- Benchmark (SPX or sector ETF or peer median).
- Universe (median of universe at same date).

Outputs: {`horizon`, `return_decision`, `return_alternative`, `vs_benchmark`, `vs_universe`}.

K.3 Decision-quality scoring (calibration)

Rolling metrics:

- **Hit rate by action band:** BUY-MEDIUM hit rate, BUY-HIGH hit rate, TRIM hit rate, SELL hit rate. "Hit" = decision outperformed alternative on chosen horizon.
- **Confusion matrix:** rows = action emitted, cols = post-hoc "should have been" given outcomes. Useful for systematic biases.
- **Brier score** for conviction bands as probabilistic claims (loose calibration; treat as informational).

- **False-confidence flags:** cases where HIGH conviction underperformed by $\geq X\%$ on 90d. System flags class for review.
- **Missed-opportunity tracking:** names not in universe that ran $\geq +50\%$ in 12 months → manual review of universe expansion needs.

K.4 Performance attribution (Brinson + factor)

Phase-2 module. At portfolio level:

- **Brinson-Fachler:** decompose vs benchmark into allocation + selection + interaction.
- **Multi-factor attribution:** value/quality/momentum/size/low-vol/sector contributions.
- **Idiosyncratic vs systematic:** how much return came from stock-specific bets vs factor tilts.

K.5 Replay engine

Phase-2 module. Runs historical points-in-time with strict look-ahead control:

- All facts in replay run must have `ts_published <= replay_asof`.
- Filings, prices, news, macro all snapped to history.
- Replay produces what-the-system-would-have-decided.
- Compare to actual user decisions and to benchmark.

This is *not* a true backtest (committee-LLM is non-stationary; running historical Sonnet completions is approximate). It is a discipline + sanity check. Treat outputs as illustrative.

K.6 Paper portfolio

User can opt to follow the engine's recommendations on a paper book. System tracks the paper book vs the real book vs benchmark.

K.7 Confusion matrix UI (admin / Settings tab)

- Heatmap: action × outcome.
- Drill into worst-calibrated cells to inspect cases.
- Filter by ticker, action, conviction, horizon.

K.8 Drawdown / risk evaluation

Portfolio-level drawdown tracker; daily compute:

- Current drawdown from 6m high.
- Max drawdown 1y, 3y.

- Drawdown trigger: $> X \rightarrow$ conviction floor lowered (more conservative); UI banner.

K.9 Data quality tracking

- Per-module daily availability metric.
- Per-source daily availability metric.
- Per-module suppression frequency (suppressed too often $\rightarrow$ improve gate).
- Schema drift events.

K.10 LLM hallucination tracking

- Validator rejection rate per persona.
- Top reasons for rejection.
- Repeated-rejection tickers flagged for prompt redesign.
- Inline citation match rate.

K.11 User override tracking

- Reason taxonomy: "I have additional context" / "I disagree with the model" / "Tax timing" / "Behavioral" / "Other".
- Aggregate reasons surfaced as feedback to the user (and to me, the architect, as design feedback).

K.12 Caveats explicitly required

- Do not present hit rates as guarantees.
 - Do not extrapolate causality without long horizons.
 - Do not show backtest curves that imply prediction power.
 - Do show calibration over time and let the user judge trust.
-

L. Unlimited-budget ("Godmode") version

The aspirational design assuming budget, time, and infrastructure are not constraints. Translate to "buildable now" in section M.

L.1 Data vendors

- **FactSet/Refinitiv/Bloomberg-grade fundamentals** (replace yfinance silent-NaN).
- **AlphaSense** for transcripts + filings full-text search with citations.
- **Aiera** live earnings calls + summaries with timestamped quotes.
- **S&P Capital IQ Compustat** for clean fundamentals + estimates.

- **Quandl/Tiingo** for alt-data layered in (insider, options, sentiment-with-source).
- **Polygon paid** for tick-level price data with options.
- **Sentio / Visible Alpha** for estimate revisions + estimate dispersion.

L.2 Background jobs / infrastructure

- **Real-time streaming** of filings (EDGAR push) into source-pack updates.
- **Distributed job queue** (Redis Streams or NATS) for module fan-out.
- **Vector store** (pgvector or dedicated) for source-pack semantic recall — "show me other names where management used similar guidance language".
- **Knowledge graph** (Neo4j) for entity resolution: companies, executives, board members, related-party entities, products, regulatory bodies.
- **CDC pipeline** to keep snapshot store synced with provider freshness without polling waste.

L.3 LLM committee (godmode)

- **Multiple models** per persona (Bull on Sonnet, Bear on a different family, Forensic on a model fine-tuned on filings, PM on Opus). Reduces correlated hallucinations.
- **Self-consistency**: each persona run 3 times with different seeds; majority vote.
- **Tool use**: committee can request additional facts (within universe) on demand; deterministic fetcher answers.
- **Fine-tuning** on the user's accepted/rejected decisions over time (RLHF-style with strict guardrails, only for ranking, never for fact generation).
- **Long-context retrieval**: entire 10-K + last 4 transcripts + last year of news in source pack with semantic citations.

L.4 Source citation engine

- **Verified citations**: every claim in PM note hyperlinks to a passage of a filing or a transcript with offsets; the user can click to read.
- **Citation provenance graph**: facts linked across sources for cross-validation.

L.5 Market scanner (Godmode radar)

- **Continuous universe scan** over the entire investable universe (US + Intl + closed-end funds + ETFs).
- **Real-time anomaly detection** on filings, insider buys, 13F deltas.
- **Alert queue** with severity ranks; user-configurable subscriptions.

L.6 Portfolio simulator + scenario engine

- **What-if simulator**: "If I add 2% to AAPL by trimming 1% from MSFT and 1% from cash, what changes for factor mix, sector mix, drawdown profile?"
- **Scenario engine**: custom macro shocks, regime changes, single-stock blowups; impact on portfolio.

L.7 Risk monitor (continuous)

- VaR + CVaR computed daily.
- Stress scenario library expanded (Asian crisis, Dotcom bust, 9/11, GFC, COVID, 2022, sector-specific).
- Drawdown alerts with auto-escalating conservatism.

L.8 Alerting

- Push notifications, email, in-app, webhook to user-configurable destinations.
- Severity-tiered: critical (filing red flag, thesis broken), important (action upgrade/downgrade), informational (catalyst window).

L.9 Backtesting / replay (godmode)

- Strict look-ahead controls.
- Multi-cycle replays (full bull, bear, sideways markets).
- LLM-frozen replay using cached committee outputs from historical runs.
- Counterfactual analysis: "if I had followed every BUY signal, what would my returns be?"

L.10 Personal investment policy (godmode)

- Tax-aware modeling with full lot-level tracking.
- Tax-loss harvesting candidate finder.
- IRA / 401(k) / brokerage account-aware.
- Estate / retirement projection integration.

L.11 Opportunity discovery (godmode)

- **Industry expert calls** transcribed and folded in (Third Bridge / Tegos).
- **Patent / regulatory / product launch** event streams.
- **Web traffic** / app download / hiring data alt-data.
- **Satellite imagery** for retail traffic / shipping / oil storage (luxury optional).

L.12 Deploy integration (godmode)

- Real-time broker integration to execute the Deploy plan.
- Pre-trade compliance check against IPS.
- Post-trade attribution feedback loop.

L.13 Performance feedback (godmode)

- Live attribution computed daily.
- Long-horizon study (5y+) of decision quality with statistical confidence.
- Behavioral metrics: trades-per-month, hold-period distribution, churn rate, vs benchmark.

L.14 Product UI (godmode)

- Multi-device (web, mobile native, watch).
- Voice queries to PM ("What's the latest on NVDA?").
- Conversational interrogation ("Why did you change to TRIM on META?") with cited answer.
- Personal CIO dashboard with health-of-portfolio score.
- Annual "investment letter" auto-drafted with citations for the user to edit.

The Godmode version is a 5–10 year roadmap, not a near-term plan.

M. Buildable-now version (realistic sequence on Next.js + FastAPI + Supabase)

The realistic 9–14 PR sequence over ~10–14 weeks delivering the core v3 promise on the existing stack with free-tier providers.

M.1 Constraints we accept

- yfinance for fundamentals + prices (with sanitizer + Finnhub fallback).
- Finnhub free tier for news + calendar.
- FRED for macro (free with API key).
- EDGAR direct for filings + 13F + Form 4 (free; 10 req/s polite).
- ETF issuer JSONs for holdings (free).
- CoinGecko for crypto.
- Supabase JSONB for snapshots (proven).
- FastAPI BackgroundTasks for run orchestration (proven).
- Anthropic Claude Sonnet/Opus mix for LLM committee (existing integration).
- Vercel for frontend (existing).
- Railway for backend (existing).

M.2 Buildable scope

- Data Truth Layer + Snapshot Store v2.
- All Layer 2 modules with free-tier sources (gates publish/suppress honestly).
- Portfolio Governor with IPS Layer 0.
- Decision Policy v1 (full state machine).
- Source Pack Builder + LLM Committee (express + standard tiers; full deferred).
- Opportunity Radar with closed universe, deterministic ranking, top-15 committee gate.
- Frontend v3 with all sections.
- Intel→Deploy typed handoff.
- Delta engine + What-Changed UI.
- Decision logs + post-decision performance windows.
- Behavioral guardrails (basic flagging).
- Thesis tracker (basic; LLM-drafted, user-approved).

M.3 Deferred to Phase 2

- `risk_monitor_v1` full stress library.
- `factor_exposure_v1` (will ship as informational chip; full attribution Phase 2).
- `replay_v1`.
- Brinson + factor attribution.
- Tax-aware turnover.
- Estimate revisions module (paid tier).
- Earnings transcripts (paid tier).
- Real-time alerting (push/email).
- Voice / conversational interrogation.

M.4 Phase plan

- **Phase 0 (1 week):** PR 0 — UI contract lock + scaffolding + canary.
 - **Phase 1 (4–6 weeks):** Data Truth + 6 axis modules + Decision Policy + Frontend v3.
 - **Phase 2 (3–4 weeks):** Source Pack Builder + LLM Committee + Opportunity Radar + Frontend integration.
 - **Phase 3 (2–3 weeks):** Behavioral guardrails + Thesis tracker + Delta engine + What-Changed UI.
 - **Phase 4 (2–3 weeks):** Intel→Deploy handoff + Decision logs + post-decision performance + Settings IPS UI.
-

N. The "Do not build" list (explicit rejections)

These are not oversights. They are conscious rejections.

1. **Fake price targets.** No "\$185 by year-end". No 12-month price targets unless explicit scenario with sourced assumptions and labeled "scenario, not target".
 2. **Unsupported predictions.** No "Stock will go up". No "We expect a 20% rally". Probabilities, not certainties.
 3. **Visible famous-investor cosplay.** No "Buffett would buy this" framing. 13F facts are evidence in the drawer, not headlines.
 4. **Broker-label confusion.** No Strong-Buy/Buy/Hold/Sell/Strong-Sell scale. Just BUY/HOLD/TRIM/SELL.
 5. **Posture buckets.** No Watchlist / Risk Watch / Add Candidate / Review labels in the visible UI.
 6. **Giant report pages.** No 30-section analyst memo dumped on screen. Drawer with progressive disclosure.
 7. **Raw metric jargon.** No `roic_ttm`, `ev_ebitda`, `fcf_margin` in the UI. Plain English only.
 8. **Noisy social sentiment as core signal.** No Reddit/Twitter aggregation as a primary recommendation driver. Source-quality-weighted news at most; demoted by default.
 9. **Over-trading triggers.** No "act now" CTAs. No daily rebalance prompts. No churn-inducing alerts.
 10. **LLM-only stock picking.** No path where an LLM emits a ticker and an action without deterministic gates, citations, validators, and the closed universe.
 11. **Technical-analysis-only decisions.** Technicals confirm or deny fundamentals; they never drive a decision alone.
 12. **Black-box scores.** No single composite "Intel score" of 0–100. Multi-axis bands only.
 13. **Fake precision.** No "94.6% confidence". Bands {LOW, MEDIUM, HIGH}.
 14. **Cross-Intel/Deploy blurring.** Intel never owns allocation math. Deploy never invents thesis text.
 15. **Continuous re-recommendation noise.** Deterministic snapshots are the unit of truth; transient between-run wobble does not surface.
 16. **Auto-execution.** Engine never auto-trades. User confirms.
 17. **Tax-naive trim/sell suggestions** (Phase 2 — until tax module exists, TRIM/SELL output explicitly states "tax considerations not yet modeled").
 18. **Universe expansion via LLM.** LLM cannot suggest tickers outside `universe_v1`.
-

O. Migration from current Intel v2 (decisive)

O.1 Delete

- The visible posture-bucket UI surfaces. The fields stay temporarily (PR 0 already done in current branch state); the visible filters and badges go to BUY/HOLD/TRIM/SELL only.
- The blanket `insufficient_data` global gate that produces HOLD-34. Replace with per-axis suppression in `decision_policy_v1`.
- The fallback-blending logic in `_build_intel_read_for_card` that mixes runs. Snapshots become per-run immutable.

O.2 Keep (internal-only)

- `_thesis_v2` and `_reasoning_v2` as transitional persistence locations. New writes go to `intel_run_snapshots`. Reads dual-source for one release.
- `thesis_engine`, `score_schema`, `reasoning_v2_builder`, `reasoning_v2_plain_english`, `thesis_plain_english` modules — these contain useful logic that v3 modules can extend; they remain backend-only and feed source-pack via adapters.
- Existing agent orchestrator (`orchestrator.py`) — extended to call new v3 modules.
- Existing `decision_logs*` tables — they become the audit trail for Layer 8 feedback.

O.3 Make internal-only

- `intel_posture_label`, `intel_filter_bucket` — keep emitting from backend for one release; UI does not consume; remove in PR 7.
- `thesis_plain_english` Business Read — keep generating; UI never renders. Drawer Forensic / Valuation persona content replaces it once `committee_v1` is live.

O.4 How to reset the UI contract

- Lock filter tabs to ALL/BUY/HOLD/TRIM/SELL via regression test. PR 0 already establishes this.
- Lock card badge to {BUY, HOLD, TRIM, SELL, WATCH (radar), AVOID (radar)} via regression test.
- Lock action set as enum in TypeScript; new labels require typecheck change AND spec update.
- Add HOLD-34 canary CI test.

O.5 How to avoid another posture-label mess

- New label categories require: 1. A spec change in this document. 2. A new typed enum value in the API contract. 3. A new regression test. 4. A migration entry in HANDOFF.
- No more "rename and ship" PRs.
- "What about edge case X?" → propose new card *modifier* (chip), not a new bucket.

O.6 Sequencing with minimal breakage

- Phase 0: lock contract + scaffold + canary. Ship.
- Phase 1: build modules behind a feature flag `intel_v3_enabled=false`. Ship in pieces.
- Phase 1 completion: turn `intel_v3_enabled=true` for the user's account; v3 cards displayed; v2 cards remain read-only behind a "legacy" toggle for 1 cycle; remove v2 path after parity confirmed.

O.7 Schema migration (Supabase)

- New table `intel_run_snapshots(run_id, ts, modules_present, content_hash, schema_version, partial, missing_modules)`.

- New table `intel_module_outputs(run_id, ticker, module_id, payload JSONB, suppressed, schema_version)`.
 - New table `policy_versions(version, payload JSONB, ts)`.
 - New table `theses(ticker, version, must_be_true JSONB, would_change_view JSONB, status, last_review_ts)`.
 - New table `decisions_v3(decision_id, run_id, ticker, action, conviction, blockers JSONB, vetoes JSONB, source_pack_hash, policy_version, ts)`.
 - All migrations additive. No existing v2 table touched in PR 0–4. PR 5+ may add columns to existing tables but never drop.
-

P. Full PR roadmap

19 PRs. Each one model, one chat, one PR, one merge gate. "Stop after PR" rule applied. Order optimized for incremental value and minimal v2 breakage.

Notation key

- **Model:** Codex / Sonnet / Opus.
 - **Chat:** new chat or follow-up.
 - **Usage:** Low / Medium / High (token + time).
 - **Risk:** Low / Medium / High.
 - **Backend / Frontend scope:** which side touched.
 - **Tests:** named test groups.
 - **Acceptance:** crisp criteria.
 - **Rollback:** how to revert.
-

PR 0 — Lock UI contract + scaffold v3 modules

- **Objective:** Lock filter tabs to ALL/BUY/HOLD/TRIM/SELL. Lock card badge enum. Add HOLD-34 canary test. Scaffold v3 module folder.
- **Files:** `v2/frontend/src/app/dashboard/recommendations/page.tsx`, `v2/frontend/src/components/cards/AgentInsightCard.tsx`, `v2/backend/app/services/intelligence/v3/__init__.py` + ~25 stub files, `v2/backend/tests/test_hold_collapse_canary.py` (new).
- **Backend / Frontend:** Both, mostly contracts.
- **Model:** Codex.
- **Chat:** New.
- **Usage:** Low.

- **Risk:** Low.
 - **Tests:** filter contract regression; badge enum regression; canary; no posture badge regression.
 - **Docs:** HANDOFF entry.
 - **Acceptance:** All filters locked; canary in CI; module stubs exist; no behavior change.
 - **What not to do:** No business logic in v3 stubs. No backend changes. No SQL.
 - **Rollback:** revert PR; canary is purely additive.
-

PR 1 — Data Truth Layer + Snapshot Store v2 + Schema migration

- **Objective:** Implement `data_truth_v1` adapter contract (citations + freshness on every fact). Implement `snapshot_store_v2` with new Supabase tables. Dual-write reasoning into both old `agent_runs.allocation` and new `intel_run_snapshots` for one release.
 - **Files:** `v2/backend/app/services/intelligence/v3/data_truth.py` (new), `snapshot_store_v2.py` (new), provider adapter additions for citation metadata, new SQL migration `016_intel_run_snapshots.sql`, orchestrator wiring.
 - **Backend:** Yes. **Frontend:** No.
 - **Model:** Sonnet.
 - **Chat:** New.
 - **Usage:** Medium.
 - **Risk:** Medium (snapshot mechanism).
 - **Tests:** citation completeness, TTL, conflict resolution, dual-write integrity, schema-drift alerts, partial snapshot semantics.
 - **Docs:** HANDOFF; SQL note in PR summary.
 - **Acceptance:** Every fact carries `{value, source, ts_published, source_url, freshness}`; new tables receive every run; v2 reads still work.
 - **What not to do:** No UI changes; no action logic; no removal of v2 reads.
 - **Rollback:** Set `dual_write=false`; snapshot tables become orphan rows; v2 path unchanged.
-

PR 2 — IPS Layer 0 + Policy versioning

- **Objective:** Implement `policy_v1` Investment Constitution. Seed with sensible defaults derived from current portfolio + handoff observations. Persist in `policy_versions`. Backend reads policy as preconditions.
- **Files:** `v2/backend/app/services/intelligence/v3/policy_v1.py` (new), SQL migration `017_policy_versions.sql`, orchestrator hook to load active policy.
- **Backend:** Yes. **Frontend:** No yet (Settings UI in PR 13).
- **Model:** Codex (mechanical seed) → Sonnet only if seeding logic non-trivial.
- **Chat:** New.

- **Usage:** Low.
 - **Risk:** Low.
 - **Tests:** policy load/persist; default seed; version increment.
 - **Acceptance:** Active policy retrievable; backend reads it before any module.
 - **What not to do:** No UI editing yet.
 - **Rollback:** Drop policy table; modules degrade to defaults.
-

PR 3 — Quality / Moat + Return Drivers + Valuation Context + Peer Set + Fund Intel

- **Objective:** Five attractiveness/valuation modules. Deterministic. Suppress honestly.
 - **Files:** v2/backend/app/services/intelligence/v3/quality_moat_v1.py, return_drivers_v1.py, valuation_context_v1.py, peer_set_v1.py, fund_intel_v1.py. ETF issuer adapter additions.
 - **Backend:** Yes. **Frontend:** No.
 - **Model:** Sonnet.
 - **Chat:** New.
 - **Usage:** Medium-High.
 - **Risk:** Medium.
 - **Tests:** ≥3-metric gates, ETF/crypto routing, no-fake-peer-medians, ROIC>WACC enforcement, valuation band logic.
 - **Acceptance:** Modules emit valid outputs for the held universe; suppression visible for thin coverage.
 - **What not to do:** No UI; no decision-policy wiring yet.
 - **Rollback:** Modules untouched by readers; safe to leave dormant.
-

PR 4 — Earnings Quality + Filing Risk + Dilution Risk + Cashflow Quality + Balance Sheet Resilience

- **Objective:** Five forensic / risk modules.
- **Files:** earnings_quality_v1.py, filing_risk_v1.py, dilution_risk_v1.py, cashflow_quality_v1.py, balance_sheet_resilience_v1.py. EDGAR adapter additions for share-count history, going-concern detection, late-filing flags, RPT extraction.
- **Backend:** Yes. **Frontend:** No.
- **Model:** Sonnet.
- **Chat:** New.
- **Usage:** High.
- **Risk:** Medium-High (EDGAR rate limits, parsing edge cases).
- **Tests:** Beneish-M, Sloan accruals, cash-conversion math, going-concern detector, late-filing flag, auditor-change flag, RPT threshold, dilution detection, SBC% math, foreign-ticker suppression, known-clean and known-flagged

fixtures.

- **Acceptance:** Modules surface known flags on test fixtures; suppression honest. **Stop after PR.**
 - **Rollback:** Modules dormant if not consumed.
-

PR 5 — Macro Regime + Technical Confirmation + Catalyst Watch + Sentiment Filter + Short/Options Tape

- **Objective:** Five modifier / timing modules.
 - **Files:** macro_regime_v1.py, technical_confirmation_v1.py, catalyst_watch_v1.py, sentiment_noise_filter_v1.py, short_options_tape_v1.py. FRED adapter (new).
 - **Backend:** Yes. **Frontend:** No.
 - **Model:** Sonnet.
 - **Chat:** New.
 - **Usage:** Medium.
 - **Risk:** Low-Medium.
 - **Tests:** regime classification on known dates, multi-horizon momentum reproducibility, calendar accuracy, no-social-as-core, short-interest/IV-skew gates.
 - **Acceptance:** Modules emit bands; macro one regime per run.
-

PR 6 — Insider/Inst + Factor Exposure + Customer Concentration + Regulatory/Litigation + Geopolitical

- **Objective:** Five evidence-rich modules.
 - **Files:** insider_inst_v1.py, factor_exposure_v1.py, customer_concentration_v1.py, regulatory_litigation_risk_v1.py, geopolitical_supply_chain_v1.py. EDGAR Form 4 + 13F parsing.
 - **Backend:** Yes. **Frontend:** No.
 - **Model:** Sonnet.
 - **Chat:** New.
 - **Usage:** Medium-High.
 - **Risk:** Medium.
 - **Tests:** insider cluster detection, 13F delta math, factor-loading sanity, RPT detection.
 - **Acceptance:** Modules emit valid bands.
-

PR 7 — Portfolio Governor + Decision Policy Engine

- **Objective:** Implement `portfolio_governor_v1` with caps and capital priorities; implement `decision_policy_v1` full state machine. Replace the per-ticker badge logic. New `recommendation_engine_v3` reads v3 snapshots and projects to a v3 InsightCard adapter (still consumed by v2 UI for now).
 - **Files:** `portfolio_governor_v1.py`, `decision_policy_v1.py`, `recommendation_engine_v3.py`, adapter to v2 InsightCard, `behavioral_guardrails_v1` stub.
 - **Backend:** Yes. **Frontend:** Adapter only.
 - **Model:** Sonnet.
 - **Chat:** New.
 - **Usage:** High.
 - **Risk:** High (heart of v3).
 - **Tests:** Policy table-driven tests covering every action transition; HOLD-34 canary; portfolio fit math; edge cases (single-ticker, all-cash, all-ETF); thin-data per-axis suppression preserves differentiation; behavioral flags.
 - **Acceptance:** On the 34-ticker portfolio with current data, action distribution shows ≥ 3 distinct actions; canary passes. **Stop after PR.**
 - **What not to do:** No LLM committee yet; no UI changes; no Deploy contract yet.
 - **Rollback:** Toggle `decision_policy_v3_enabled=false`; v2 path resumes.
-

PR 8 — Source Pack Builder + LLM Committee v1 (dark launch)

- **Objective:** Implement `source_pack_builder` with typed schema; implement personas Bull/Bear/Forensic/Valuation/Technical/Risk/PM with strict validators; implement `committee_validator`. **Persist** committee output but do NOT render in UI. Cost cap algorithm in place.
 - **Files:** `source_pack_builder.py`, `committee_v1.py`, `committee_validator.py`, `committee_personas/*.py` prompts, persistence in `intel_run_snapshots`.
 - **Backend:** Yes. **Frontend:** No.
 - **Model:** Sonnet (multi-persona prompting requires care).
 - **Chat:** New.
 - **Usage:** High.
 - **Risk:** High.
 - **Tests:** Validator rejection cases (every forbidden phrase, every uncited claim, every raw key, every fabricated number, every out-of-universe ticker), schema conformance, source-pack hash caching, cost cap enforcement, citation resolution rate.
 - **Acceptance:** For ≥ 10 sample tickers, committee produces validated output; validator rejects 100% of injected hallucinations on a curated test set; cost-per-run within budget. **Stop after PR.**
 - **Rollback:** Toggle `llm_committee_enabled=false`; deterministic decisions still produced.
-

PR 9 — Universe Builder + Opportunity Radar (dark launch)

- **Objective:** Implement `universe_v1` with closed-universe loaders; implement `opportunity_radar_v1` with screens, deterministic ranking, top-15 committee gating. **Persist** radar output but do NOT render in UI yet.
 - **Files:** `universe_v1.py`, `opportunity_radar_v1.py`, `screens` (`screens/*.py`), `persistence`.
 - **Backend:** Yes. **Frontend:** No.
 - **Model:** Sonnet.
 - **Chat:** New.
 - **Usage:** High.
 - **Risk:** High (universe + LLM cost).
 - **Tests:** Universe gating (no out-of-universe), dedup vs holdings, radar evidence gates, ranking determinism, max-results, screen quotas.
 - **Acceptance:** Radar surfaces ≥ 10 candidates per run on broad portfolios; cost cap respected. **Stop after PR.**
 - **Rollback:** Toggle `radar_enabled=false`.
-

PR 10 — Frontend v3 page shell + IntelCardV3 + filter tabs + drawer

- **Objective:** Build the new page structure with all sections. Switch held-cards display to `recommendation_engine_v3` output. Drawer renders deterministic axes (Bull/Bear from committee in PR 11).
 - **Files:** `v2/frontend/src/app/dashboard/recommendations/page.tsx`, `IntelCardV3.tsx` (new), `TickerDetailDrawer.tsx` (new), `lib/api.ts` (new types).
 - **Backend:** Adapter touch only. **Frontend:** Major.
 - **Model:** Sonnet under UI budget.
 - **Chat:** New.
 - **Usage:** High.
 - **Risk:** High.
 - **UI budget:** Phase = recommendations; Max files = 8; Forbidden surfaces = Deploy / Positions / Deposits / DRIP / Import; Stop = 8 files; Decision = APPROVE.
 - **Tests:** Visual regression, filter contract, no raw-key leak, drawer open/close, suppression UI.
 - **Acceptance:** Card distribution shows ≥ 3 distinct actions; drawer renders deterministic content. **Stop after PR.**
-

PR 11 — Frontend integrates LLM committee + Bull/Bear drawer + What-Changed

- **Objective:** Render committee output (Bull/Bear/Forensic/Valuation/PM note + dissent) in drawer; render delta engine output in What-Changed section; add "Committee deferred" badge for cost-capped tickers.
- **Files:** `IntelCardV3.tsx` extension, `whatChanged.tsx` (new), `delta_engine_v1.py` integration.
- **Backend:** `delta_engine` wiring. **Frontend:** Major.
- **Model:** Sonnet.
- **Chat:** New.

- **Usage:** Medium.
 - **Risk:** Medium-High.
 - **Tests:** Committee render with citations; missing-committee fallback; what-changed delta diffs.
 - **Acceptance:** Committee output rendered cleanly with citations.
-

PR 12 — Frontend Opportunity Radar section + Best Buys + Trim/Sell Desk

- **Objective:** Render the three top-level sections; implement "from screen" tags; "show more" expand; AVOID toggle; empty states with affirmative copy.
 - **Files:** `OpportunityRadarSection.tsx`, `BestBuysSection.tsx`, `TrimSellDeskSection.tsx`.
 - **Backend:** No new logic. **Frontend:** Major.
 - **Model:** Sonnet under UI budget.
 - **Chat:** New.
 - **Usage:** Medium.
 - **Risk:** Medium.
 - **UI budget:** Max files 5; Forbidden = Deploy.
 - **Tests:** Section rendering, empty states, screen-tag display, AVOID toggle.
 - **Acceptance:** All sections functional; empty states friendly.
-

PR 13 — Settings IPS UI + behavioral guardrails surfacing

- **Objective:** Add a Settings tab where the user can edit IPS (goals, time horizon, risk tolerance, caps, sell discipline, override rules). Add second-look prompts for behavioral flags before "Use AI plan".
 - **Files:** `v2/frontend/src/app/dashboard/settings/page.tsx` (new), policy edit UI, integration with Deploy step.
 - **Backend:** policy edit endpoints.
 - **Model:** Sonnet under UI budget.
 - **Chat:** New.
 - **Usage:** Medium.
 - **Risk:** Medium.
 - **UI budget:** Max files 6; Forbidden = Recommendations / Deploy / Position / Import / Deposits.
 - **Tests:** Policy load/save; second-look prompt logic.
 - **Acceptance:** User can edit policy; behavioral flags surface in Deploy.
-

PR 14 — Intel→Deploy typed handoff

- **Objective:** Define and ship IntelToDeployHandoff schema; refactor Deploy to read only from this contract.

- **Schema:**

```
IntelToDeployHandoff = {
  asof: ts,
  policy_version: str,
  tickers: [{
    ticker: str,
    action: BUY|HOLD|TRIM|SELL|WATCH|AVOID,
    conviction_band: HIGH|MEDIUM|LOW,
    capital_priority: urgent_buy|opportunistic_buy|hold|trim_first|sell_first,
    max_position_pct: float,
    must_not_exceed_pct: float,
    blockers: [str],
    suppression_reasons: [str],
    why_now: str | null
  }],
  portfolio_context: {
    cash_pct: float,
    risk_state: str,
    recent_drawdown: float
  }
}
```

- **Files:** recommendation_engine_v3.py (handoff emit), Deploy router input adapter, lib/api.ts types.
 - **Backend:** Yes. **Frontend:** Type wiring only.
 - **Model:** Codex (mechanical).
 - **Chat:** New.
 - **Usage:** Low-Medium.
 - **Risk:** Medium (don't break Deploy).
 - **Tests:** Handoff schema validation, Deploy reads work end-to-end, allocation math unchanged.
 - **Acceptance:** Deploy receives same downstream behavior via the new contract.
-

PR 15 — Decision logs v3 + post-decision performance windows

- **Objective:** Implement decisions_v3 table; capture every decision the policy emits with full provenance. Add post-decision performance windows (7/30/90/180/365d) computed daily by a cron.
 - **Files:** SQL 018_decisions_v3.sql, feedback_v1.py (basic windows), cron job.
 - **Backend:** Yes. **Frontend:** Drawer "performance feedback" section (collapsed by default).
 - **Model:** Sonnet.
 - **Chat:** New.
 - **Usage:** Medium.
 - **Risk:** Medium.
 - **Tests:** Decision capture, window math, benchmark alignment.
 - **Acceptance:** Past decisions appear in drawer with horizon returns.
-

PR 16 — Thesis tracker + drift detection + What-Changed thesis section

- **Objective:** Implement `thesis_tracker_v1`; LLM drafts thesis on first encounter; user approves; drift engine flags changes.
 - **Files:** `thesis_tracker_v1.py`, `SQL 019_theses.sql`, drawer "Thesis check" UI, What-Changed drift section.
 - **Backend:** Yes. **Frontend:** Drawer addition.
 - **Model:** Sonnet.
 - **Chat:** New.
 - **Usage:** Medium.
 - **Risk:** Medium.
 - **Tests:** Drift detection on canonical examples; broken-thesis detection.
 - **Acceptance:** Each held position has a thesis; drift surfaces in What-Changed.
-

PR 17 — Risk Monitor (basic stress + concentration alerts)

- **Objective:** Implement `risk_monitor_v1` with 2008/2020/2022 replays + rate/USD/oil shocks + concentration cluster detection. Surface portfolio-level "stress" chip.
 - **Files:** `risk_monitor_v1.py`, Portfolio Command Center stress chip.
 - **Backend:** Yes. **Frontend:** Chip + drawer expansion.
 - **Model:** Sonnet.
 - **Chat:** New.
 - **Usage:** Medium.
 - **Risk:** Medium.
 - **Tests:** Replay accuracy on known periods; concentration detection.
 - **Acceptance:** Chip renders sane stress numbers.
-

PR 18 — Cleanup + remove v2 dual-write + remove deprecated fields

- **Objective:** After 1–2 cycles of stable v3, remove `_thesis_v2` / `_reasoning_v2` writes from `agent_runs.allocation`; remove `intel_posture_label` / `intel_filter_bucket` fields; remove unused legacy code paths in `_compute_insight_cards`.
- **Files:** Cleanup throughout backend + frontend types.
- **Backend:** Yes. **Frontend:** Type cleanup.
- **Model:** Codex.
- **Chat:** New.
- **Usage:** Low-Medium.

- **Risk:** Medium.
 - **Tests:** Existing tests must still pass; deletion of unused functions verified.
 - **Acceptance:** Backend code reduced; v2 fields gone.
 - **Rollback:** Revert PR if any consumer broken.
-

PR 19 — Replay v1 (admin tooling)

- **Objective:** Implement `replay_v1` for historical points-in-time replays with strict look-ahead controls. Admin/Settings tab to run replays.
 - **Files:** `replay_v1.py`, admin UI in Settings.
 - **Backend:** Yes. **Frontend:** Settings sub-tab.
 - **Model:** Sonnet.
 - **Chat:** New.
 - **Usage:** Medium-High.
 - **Risk:** Medium-High (look-ahead bias).
 - **Tests:** Look-ahead detection; replay determinism.
 - **Acceptance:** Replay produces deterministic outputs on canonical dates.
-

Optional later PRs (Phase 4+)

- PR 20: Brinson + factor attribution dashboard.
 - PR 21: Tax-aware turnover module.
 - PR 22: Estimate revisions module (paid tier).
 - PR 23: Real-time alerting (push/email).
 - PR 24: Earnings transcripts integration (paid tier or curated press releases).
-

Q. Security and compliance safety

Even though this is a personal app, treat truth-handling rigorously.

Q.1 Disclaimers

- Every page footer: "This tool surfaces information; it does not provide financial advice. Decisions are yours."
- Drawer: "Numbers shown are sourced from {provider} as of {ts}. Free-tier data may have gaps; do not assume institutional quality."
- Settings: An on-by-default disclosure read once at first run.

Q.2 No guaranteed advice

- No language implying certainty.
- No "recommended" without "subject to your own due diligence".
- No "guaranteed" anywhere.

Q.3 Audit logs

- Every decision, every override, every policy edit is logged with timestamp + actor (user|system|cron).
- Logs immutable for 1 year.
- Settings tab can export logs.

Q.4 Sources never fabricated

- Validators reject unsourced claims.
- Adapter health logs schema drift.
- Citations link out to source URLs (open in new tab).

Q.5 Distinguish facts, model judgments, and user preferences

- UI styling differentiates:
 - **Facts:** plain text with hover citation.
 - **Model judgments:** styled with subtle "AI" or "Model" tag.
 - **User preferences:** styled with subtle "Your policy" tag.
- This is the single most important trust feature: the user always knows whether a sentence is a fact, a judgment, or their own setting reflected back.

Q.6 Data privacy

- All credentials in env; never in logs.
- Supabase RLS enforced on user tables.
- LLM payloads scrub PII (none expected; principle).
- No telemetry to third parties beyond LLM calls.

Q.7 Compliance posture

- Single-user / personal app: SEC IA Act not implicated.
- If ever shared, become "investment information" under state law — disclaimers sufficient for personal use; productize requires legal review.

R. Known unknowns and explicit unknowns

This is a deliberately humble section. Top of the unknown list:

1. **How well will the LLM committee perform on edge cases (tiny caps, weird fund structures, recent IPOs).** Validator is aggressive; expect first 3 weeks of false rejections; iterate prompts.
 2. **How sensitive will the decision policy be to band thresholds.** First version is best-guess from research synthesis; calibrate from feedback layer over time.
 3. **Whether the universe of 3000 names is too large or too small.** Telemetry: count of LLM-rejected suggestions, count of user-flagged "implausible" radar entries.
 4. **How aggressive to be about behavioral guardrails before they annoy.** Start as soft warnings; escalate only if user opts in.
 5. **Whether forensic accounting modules false-positive on financials/REITs.** Industry-aware thresholds; but coverage gaps probable.
 6. **EDGAR rate-limit pressure under full radar.** Token-bucket conservative at 6 req/s sustained; escalate only if validation passes.
 7. **Whether IPS defaults match the user's actual preferences.** First-run wizard nudges user to confirm; hard caps configurable.
-

S. First three copyable implementation prompts

The first three prompts to paste into Claude/Codex after this spec is accepted. Each is scoped to one PR. Each prompt is self-contained and follows the repo's PROMPT_LIBRARY conventions.

S.1 — PR 0 prompt (UI contract lock + scaffold)

Model: Codex **Chat:** New chat **Usage estimate:** Low **Why:** PR 0 is mechanical — lock visible UI contract, scaffold v3 module folder, add HOLD-34 canary. No new logic.

Task:

Lock Intel v3 UI contract foundations + scaffold v3 module folder. No behavior change for users.

Severity classification: Level 1.

Assumptions:

- Visible Intel UI is BUY/HOLD/TRIM/SELL only (already restored). Posture buckets are out of the visible UI permanent.
- v3 module work begins next PR; this PR ships scaffolding and regression tests only.
- Backend behavior unchanged.

Success criteria:

- Filter tabs are locked and protected by a regression test.
- Card badge enum is locked to {BUY, HOLD, TRIM, SELL, WATCH, AVOID} and protected by a regression test.
- A HOLD-34 canary test exists and fails if a recommendation set ever yields all HOLD/LOW for a non-degenerate snapshot.
- v3 module folder exists with stub files.

- HANDOFF.md updated with PR 0 entry.
- No user-visible behavior change.

Repo-local files to read FIRST (smallest necessary subset):

- CLAUDE.md
- docs/ai/HANDOFF.md (most recent entry only)
- docs/ai/PROMPT_LIBRARY.md (only "Prompt reviewer gate", "Discovery budget rule")
- docs/ai/EXECUTION_PRINCIPLES.md
- docs/ai/skills/implementation.md
- v2/frontend/src/app/dashboard/recommendations/page.tsx
- v2/frontend/src/components/cards/AgentInsightCard.tsx
- v2/frontend/src/components/cards/AgentInsightCardThesisVisibility.test.tsx

Execution principles: before coding, state assumptions and success criteria; keep changes simple and surgical; ever

Discovery budget:

- Do not run find/grep/glob.
- Read primary edit targets first.
- No fallback reads unless blocked.

Exact scope:

1. In v2/frontend/src/app/dashboard/recommendations/page.tsx, add a contract-lock comment above INTEL_FILTERS that
2. In v2/frontend/src/components/cards/AgentInsightCard.tsx, lock the badge contract: only {BUY, HOLD, TRIM, SELL,
3. Create v2/backend/app/services/intelligence/v3/ folder with __init__.py and 25 stub files (one docstring each, n
4. Add v2/backend/tests/test_hold_collapse_canary.py: integration test that fails when a recommendation_engine resp

Constraints:

- No backend logic changes.
- No new dependencies.
- No SQL.
- No Deploy changes.
- No HANDOFF entry beyond a 5-line PR 0 summary.

Tests:

- Add tests as specified above.
- Run focused tests for changed files only.

Docs:

- Update docs/ai/HANDOFF.md with: PR 0 lock + scaffold entry, files changed, tests, no-Supabase-SQL, no-README-edit

MANDATORY:

- Update docs/ai/HANDOFF.md in the same change.
- After opening the PR, stop. Do not propose the next implementation prompt.

Final report format:

Severity classification: Level 1

Root cause/plan:

Files changed:

Tests:

Risks:

Supabase SQL: No

HANDOFF.md edited: Yes

README.md edited: No

S.2 — PR 1 prompt (Data Truth Layer + Snapshot Store v2 + dual-write)

Model: Sonnet **Chat:** New chat **Usage estimate:** Medium **Why:** Multi-file deterministic logic touching adapters, new SQL, and orchestrator wiring.

Task:

Implement Intel v3 Data Truth Layer (data_truth_v1) and Snapshot Store v2 (snapshot_store_v2). Add Supabase tables

Severity classification: Level 2 (full plumbing across adapters, store, orchestrator).

Assumptions:

- The v3 module folder exists from PR 0 with stubs.
- Existing yfinance and finnhub adapters return raw payloads without citation metadata.
- Snapshots today live implicitly in agent_runs.allocation JSONB. v3 needs explicit per-run, per-ticker, per-module
- Dual-write is a transitional safety; reads from v3 path will start in PR 7 (recommendation_engine_v3).
- This PR ships ZERO UI changes and zero action-policy changes.

Success criteria:

- Every TruthRow emitted by adapters carries {value, unit, source, ts_published, ts_observed, source_url, freshness}
- New tables intel_run_snapshots and intel_module_outputs accept writes; a content_hash is recorded per snapshot.
- A run can be retrieved end-to-end from the new tables.
- Existing v2 reads (recommendation_engine, agent_insights, etc.) still work.
- Schema-drift detector logs alerts when an adapter response shape changes.
- Per-fact TTLs enforced; expired facts emit suppressed=true rows.
- Conflict resolution between two providers logged.

Repo-local files to read FIRST (smallest necessary subset):

- CLAUDE.md
- docs/ai/HANDOFF.md (latest 2 entries)
- docs/ai/PROMPT_LIBRARY.md (sections: Discovery budget rule, Timeout / continue budget rule)
- docs/ai/EXECUTION_PRINCIPLES.md
- docs/ai/skills/supabase_change.md
- docs/ai/skills/implementation.md
- v2/backend/app/services/agents/data_sources.py
- v2/backend/app/services/agents/orchestrator.py
- v2/backend/app/services/intelligence/snapshot_store.py
- v2/database/ (look at most recent migration to mirror style)
- v2/backend/app/services/intelligence/v3/data_truth.py (stub)
- v2/backend/app/services/intelligence/v3/snapshot_store_v2.py (stub)

Execution principles: before coding, state assumptions and success criteria; keep changes simple and surgical; ever

Discovery budget:

- Do not run find/grep/glob.
- Read primary edit targets first.
- Fallback reads only if blocked, with one-sentence reason.

Timeout budget:

- If you are about to exceed time/context, stop after a checkpoint with: files changed, tests run, current status,
- Max two continues, then ask the user to move the checkpoint to a fresh chat.

Exact scope:

1. Implement v2/backend/app/services/intelligence/v3/data_truth.py:
 - TruthRow dataclass and module-level emit() helper.
 - Per-fact TTL config dict.
 - coerce_or_suppress() utility for NaN/None/inf.
 - Adapter wrappers (decorators) that bolt citation metadata onto existing yfinance / finnhub / FRED-stub / EDGAR
 - Conflict resolution: when two sources emit the same (ticker, fact, asof), prefer higher source_priority and lo
2. Implement v2/backend/app/services/intelligence/v3/snapshot_store_v2.py:
 - persist_snapshot(run_id, ticker_modules: dict) writes intel_run_snapshots row + intel_module_outputs rows.
 - read_snapshot(run_id) returns per-ticker per-module map.
 - content_hash deterministic over modules_present + payload hashes.
 - schema_version field.
3. Add SQL migration v2/database/016_intel_run_snapshots.sql with two tables and indexes.
4. Wire data_truth into adapter pipeline so existing modules continue to work but emit TruthRows.
5. Wire orchestrator (Phase 3 / Phase 4 of run) to call snapshot_store_v2.persist_snapshot(...) at end of each run,
6. Add provider_health.py (small) and a daily-callable check_health() for adapters; alerts via log.
7. Tests:
 - test_data_truth.py: citation completeness, TTL enforcement, NaN sanitizer, conflict resolution.
 - test_snapshot_store_v2.py: idempotent writes, hash determinism, partial snapshot semantics, schema_version.
 - test_orchestrator_dual_write.py: a run produces both legacy allocation writes and v3 snapshot writes; v3 reads

Constraints:

- No UI changes.
- No action-policy changes.
- No removal of v2 reads.
- No new external API dependencies; reuse existing yfinance/finnhub/EDGAR-shape calls (FRED can be a stub adapter a
- Strict Supabase RLS unchanged.

- Migration must be additive; no DROP, no ALTER on existing tables.

Tests:

- Add tests as specified.
- Run focused tests only.

Docs:

- Update docs/ai/HANDOFF.md with PR 1 entry.
- State Supabase SQL: Yes (016_intel_run_snapshots.sql).

MANDATORY:

- Update docs/ai/HANDOFF.md in the same change.
- After opening the PR, stop. Do not propose the next implementation prompt.

Final report format:

Severity classification: Level 2

Root cause/plan:

Files changed:

Tests:

Risks:

Supabase SQL: Yes

HANDOFF.md edited: Yes

README.md edited: No

S.3 — PR 2 prompt (IPS Layer 0 + Policy versioning)

Model: Codex (escalate to Sonnet only if seeding logic non-trivial) **Chat:** New chat **Usage estimate:** Low **Why:**

Mechanical setup of policy persistence + sensible defaults seeded from current portfolio state.

Task:

Implement Intel v3 Layer 0: User Investment Constitution (policy_v1) with Supabase persistence and policy versioning

Severity classification: Level 1.

Assumptions:

- PR 0 scaffolded v3/policy_v1.py.
- PR 1 introduced intel_run_snapshots; this PR adds a separate policy_versions table.
- No frontend Settings UI yet (PR 13). Backend reads the active policy as a precondition for every module call.
- Defaults can be seeded from the current 34-ticker portfolio + reasonable assumptions (e.g., 8% concentration cap,

Success criteria:

- Active policy can be read (get_active_policy()) and written (write_policy(payload)).
- Policy is versioned; each write increments version and stores prior payloads immutably.
- Default seed is created on first call if no policy exists.
- Backend modules (when added in later PRs) can read policy via a single dependency.

Repo-local files to read FIRST:

- CLAUDE.md
- docs/ai/HANDOFF.md (latest 2 entries)
- docs/ai/PROMPT_LIBRARY.md (Discovery budget rule)
- docs/ai/EXECUTION_PRINCIPLES.md
- docs/ai/skills/supabase_change.md
- v2/backend/app/services/intelligence/v3/policy_v1.py (stub)
- v2/database/016_intel_run_snapshots.sql (mirror style)

Execution principles: before coding, state assumptions and success criteria; keep changes simple and surgical; ever

Discovery budget:

- No find/grep/glob.
- Read listed files first.

Exact scope:

1. v2/backend/app/services/intelligence/v3/policy_v1.py:

- Policy dataclass with fields: goals, time_horizon, risk_tolerance, liquidity_floor_pct, sleeves[], concentrati
 - get_active_policy() reads latest from policy_versions; seeds default if empty.
 - write_policy(payload) inserts a new row with version = max+1.
 - Default seed values (named constants, well-commented).
2. SQL migration v2/database/017_policy_versions.sql:
 - policy_versions(version int PRIMARY KEY, payload jsonb, created_at timestamptz DEFAULT now(), actor text).
 3. Wire orchestrator to load active policy at run start and pass it through state to modules (read-only).
 4. Tests:
 - test_policy_v1.py: load default, write new, version increment, immutability of old versions.

Constraints:

- No UI.
- No business logic that uses policy yet (later PRs).
- Migration additive only.

Tests:

- Add tests as specified. Focused tests only.

Docs:

- Update docs/ai/HANDOFF.md.
- Supabase SQL: Yes (017_policy_versions.sql).

MANDATORY:

- Update docs/ai/HANDOFF.md in the same change.
- After opening the PR, stop. Do not propose the next implementation prompt.

Final report format:

Severity classification: Level 1

Root cause/plan:

Files changed:

Tests:

Risks:

Supabase SQL: Yes

HANDOFF.md edited: Yes

README.md edited: No

T. Final opinionated recommendation

The current Intel v2 has been patched seven times and the patches keep producing the same symptom: HOLD-34 with renamed badges. Patching it again is the lowest-yield use of any future token. Build v3.

Three opinions I'll defend:

1. **Multi-axis decomposition is the only architecture that survives sparse data.** A single weighted score will recreate HOLD-34 the day after merge. Four independent axes (evidence, attractiveness, price, fit) plus modifiers and vetoes is the minimum viable design.

1. **Source-pack first, LLM second is the only honest architecture.** Inverting the order yields hallucinated numbers that look professional. The validator has to be airtight: closed universe, citation grammar, numeric grounding, forbidden phrases, no raw keys, no fake price targets. Up to 41% of finance LLM queries hallucinate without grounding; with strict RAG and validation, that rate plummets — but the validator must be the guard, not the prompt.

1. **Opportunity Radar is what makes this an investment committee, not a tracker.** Without it, the engine can only react to the user's existing book. With it — closed universe, deterministic ranking, top-15 LLM gate — the engine actively brings the user better ideas with sourced evidence.

Three things I refuse to promise:

- Not "alpha".
- Not "right calls".
- Not certainty about anything the market reveals.

The product promise is **better-quality decisions per unit of evidence**, with brutal honesty about what is unknown. That promise is achievable and defensible. Do not let scope creep redefine it.

Three behavioral commitments to enforce on this build:

- **Stop after each PR.** Do not chain PRs in one chat. Bring the merged PR back to ChatGPT for the next prompt.
- **No new label categories without a spec change.** This document is the contract. Adding "Watch List" or "Review" or "Add Candidate" requires editing the spec first and then building it.
- **No LLM in any decision-producing path.** LLMs explain. The decision policy decides. If a future PR ever puts an LLM upstream of the action label, reject it.

Start with PR 0 today. Do not start PR 1 until PR 0 ships and the canary is green. Do not skip the "stop after PR" rule. The discipline of the rollout is itself part of the architecture.

Severity classification: Level 3 – split plan / architecture spec (Draft 2)

Root cause/plan: Intel v2 patch loop is structural; replace single-funnel decisioning with a 9-layer architecture (

Files changed: None (spec only).

Tests: None added (canary test specified for PR 0).

Risks: See section A.10 + Section R.

Supabase SQL: No (PR 1 introduces 016_intel_run_snapshots.sql; PR 2 introduces 017_policy_versions.sql; later PRs a

HANDOFF.md edited: No (this is a spec output; PR 0 will add a HANDOFF entry when shipped).

README.md edited: No.

Sources

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